A tailored exact method for static bicycle redistribution with inventory-ratio dispersion

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Abstract

Static bicycle redistribution couples attainable station inventories with capacity- and time-feasible vehicle routes. We minimize inventory-to-target ratio Gini dispersion plus weighted two-sided relative target deviation under empty departure and unique nonzero one-way station service. The sole proposed method, frozen M-B, integrates an independently paid finite startup H, inventory-state products with implicit service integrality, valid static strengthening and complete scoped interval coverage around native LP/MIP solves. We distinguish ideal validity from floating-point numerical certification. A prospectively fixed six-problem V15/V20 panel compares twelve complete M-B and cold original compact P-GRB arms under 600/1200-second per-method production caps. M-B certifies 5/6 problems (5 nonzero); P-GRB certifies 3/6 (3 nonzero). There are 3 both-certified, 2 M-B-only, 0 P-only, 1 both-open and 0 unevaluable pairs. Time ratios are reported only on the both-certified intersection and use complete normal production cost. The resulting evidence classification is limited direct certification support. Earlier six-problem V20/V35/V50 endpoint results and one known V50 long window remain separate capability-boundary evidence; their smaller gaps are not certification speedups. The results concern this fixed problem set and complete-method budget, without a universal runtime or component-causality claim.

Keywords: bicycle redistribution; inventory-to-target ratio; Gini dispersion; mixed-integer programming; valid inequalities; numerical optimality certificate.

1. Introduction

An overnight redistribution plan has two coupled tasks: choose final station inventories and realize them by feasible vehicle routes. A desired inventory vector may look balanced yet require more travel, capacity or handling time than the fleet can provide. Conversely, optimizing routes toward fixed targets can obscure the allocation decision when those targets cannot all be attained. We consider this interaction directly. A final inventory is assessed by its ratio to a station-specific target, and routes determine which combinations of ratio changes are physically attainable.

Two objective components are useful for different reasons. Dispersion of inventory-to-target ratios measures how unevenly proportional attainment is distributed across stations. Weighted relative target deviation measures the magnitude of departure from the individual targets. Dispersion alone cannot express target attainment: a uniform 50% shortage has zero dispersion. Target deviation alone permits concentrating the remaining discrepancy where its weights are small. Their weighted sum defines a precise optimization preference without claiming that it identifies customer service, revenue or socioeconomic equity. Those broader meanings require additional demand data and a justified mapping from station targets to user outcomes.

The objective also changes the computational structure. Its Gini denominator is the endogenous sum of ratios, rather than a constant conserved stock total. Station stock need not be conserved because vehicles can return loaded. The numerator couples every station pair, and the denominator couples all stations, while feasible inventory changes are supported by an ordered load-feasible fleet. This is a different interaction from minimizing travel subject to exact target restoration or demand-interval satisfaction. An inventory relaxation can explain an objective floor, but an inventory solution does not establish route feasibility.

We evaluate a complete tailored exact method, denoted M-B, for this stated problem. “Exact” identifies the formulation and complete proof obligations of the method; reported certificates remain numerical certificates under a declared floating-point policy. The method combines an independently paid feasible-solution startup with exact inventory-state products, a service representation retaining binary direction, valid strengthening and a Gini-interval controller. Gurobi provides native optimization. The controller decides which complete domains to solve or subdivide and how their returned bounds cover the original problem. A good incumbent from the startup is valuable, but is not an optimality proof.

The computational question is deliberately narrower than a claim of general superiority. Does this single integrated method complete more optimality certifications than the cold original compact formulation under matched complete production budgets, and reduce full certification cost where both certify? The fixed small-instance panel provides the direct evaluation. Earlier endpoint experiments supply an extended-range comparison. A separate longer window on one known positive-objective instance tests a proof boundary, rather than enlarging the independent sample. Both successes and losses are retained. M-B is fixed before outcomes; ENS-C is a historical predecessor/representation comparator, and no method is selected separately for each instance.

The paper has three contributions.

1. It gives a complete, source-faithful formulation and a problem-specific derivation of the inventory-state/service representation. The service quantities can be continuous without changing integer solutions under explicit uniqueness and direction premises; state/service equalities are proved valid. The physical and auxiliary Gini values are kept distinct.
2. It organizes the complete integrated algorithm and its proof contract. Objective-preserving embeddings, cutoff complementation and a maintained cover explain when local native information can support an

original-problem bound. Numerical closure, finite termination and optimality certification have separate meanings.

3. It reports a preregistered six-problem, twelve-arm small-instance optimality-certification comparison against cold P-GRB. Separate historical tables retain the six R122 problems, two fixed Seed repeats, 32 arms/48 directed pairs, all unfavorable outcomes, and the known V50 long-window proof limit. They are not pooled into one certification speedup or a larger new independent sample.

The constituent tools include standard product linearization, McCormick envelopes, multistart construction, local exchange, MIP Starts and bound pruning. We do not claim new general branch-and-cut theory for these tools. The scientific object is their particular integration for a nonseparable inventory objective and its measured behavior under one budget. Literature positioning in Section 2 limits the contribution relative to neighboring models; Appendix A makes the formulation and mathematical premises explicit. Sections 3-5 describe the problem, representations and algorithm; Section 6 presents direct certification evidence; Section 7 retains the extended historical evidence; Section 8 discusses its limits.

2. Related work and contribution position

Static bicycle rebalancing covers several materially different problems. Raviv, Tzur and Forma (Raviv et al. 2013) use station dissatisfaction and routing considerations with time resources. Their arc and sequence formulations have different service restrictions; the early full preprint inspected here must not be collapsed into one blanket single-visit model. Our objective is the specified ratio dispersion and relative target deviation, without a travel-cost objective term. Final-article metadata and the early preprint are distinguished in the literature ledger.

The publisher abstract of Chemla, Meunier and Wolfler Calvo (Chemla et al. 2013) describes a single capacitated vehicle restoring a target distribution with revisits and temporary storage. It distinguishes branch-and-cut lower bounding of a relaxation from tabu-search feasible upper bounds. We use only these abstract-supported features; its detailed initial-load and handling constraints were not obtained. Our unique one-way station service excludes both revisits and buffering, while permitting target deviations.

Erdoğan, Laporte and Wolfler Calvo (Erdoğan et al. 2014) study interval inventory requirements with one vehicle, route/handling cost and exact branch-and-cut formulations, including Benders decomposition. Their fixed-route network explicitly accounts for depot supply and handling. Erdoğan, Battarra and Wolfler Calvo (Erdoğan et al. 2015) instead address target restoration with revisits and temporary storage using an exact routing/feasibility framework. Its depot imbalance and target-equality semantics differ from empty departure and optional target attainment here. Neither problem can be adopted unchanged as a matched benchmark.

Schuijbroek, Hampshire and van Hoeve (Schuijbroek et al. 2017) connect service-level inventory intervals with multi-vehicle routing. Their routing objective is makespan, with given initial vehicle inventories and an artificial ending node; their formulation permits more service flexibility than unique one-way visits, subject to its transshipment limits. The cluster-first framework is an approximate multi-vehicle approach, alongside MIP and CP models. This provides a close allocation/routing comparison, but not the same physical model, objective or certificate scope.

The openly available Ikonen, Heljanko and Harjunkoski article (Ikonen et al. 2025) reformulates routing/load disjunctions into a MILP for demand intervals. Duplicated stations permit split visits without temporary storage; prescribed initial load and open/loop options are explicit. Its single-vehicle exact model can be combined with approximate multi-vehicle clustering. Its disjunctions and our Gini proof cover address different structures, despite shared terminology. Reported larger station counts cannot establish relative speed across the different objectives, instances, hardware and thread limits.

Its certificates concern its finite duplication model; solutions requiring additional visits or temporary operations can lie outside that generated model.

Equity requires specifying the resource being balanced. Matl, Hartl and Vidal (Matl et al. 2018) analyze workload-equity functions and consequences in vehicle routing. We use their distinction as modeling context; driver-workload conclusions are not transferred as theorems about station inventory-to-target ratios. The bibliography consistently uses the 2018 volume year and records first online publication in 2017.

The search was updated through 18 October 2026. Qiao et al. (Qiao et al. 2026) describe two-echelon coordination with task-duration fairness, an exact first-echelon subproblem and a hybrid second-echelon method. Caggiani et al. (Caggiani et al. 2026) combine free-floating electric relocation, battery swapping and equity evaluation. These are abstract/excerpt-level checks; their complete constraints remain unverified here. They establish that fairness-aware rebalancing is an active prior topic, and preclude any broad claim that introducing fairness or Gini into rebalancing is itself our novelty.

Table 1. Closest model distinctions. "Unknown" means not established by the accessed source. The three middle detailed comparisons were checked against their full model sections. Table entries do not assert interchangeable feasible sets.

Work	Objective / target requirement	Fleet and visits	Temporary or split service	Initial/return and time semantics	Exact scope
Chemla 2013	Target distribution	Single; revisits	Buffering allowed	Detailed constraints unknown from abstract	Relaxation bound plus feasible search
Erdoğan 2014	Travel/handling; hard inventory intervals	Single; at most once	No split visit	Depot stock bounds departure; depot handling charged	Full and Benders branch-and-cut
Erdoğan 2015	Travel; exact targets	Single; revisits	Temporary storage allowed	Depot imbalance participates; not our empty-departure model	Routing plus time-extended feasibility
Schuijbroek 2017	Makespan; service-level intervals	Multiple; sequence indexed	Limited transshipments; service not uniquely assigned as here	Given initial load; artificial open end; handling cost	Joint MIP/CP; clustering heuristic
Ikonen 2025	Time/distance/handling; intervals	Single model; duplicated visits	Split permitted; temporary storage forbidden	Given initial load; open or loop option	MILP per vehicle; clustered fleet approximate
Present model	Ratio Gini plus weighted relative target deviation	Multiple; one nonzero service per station	Neither split nor temporary service	Empty departure; loaded closed return with unloading; hard route horizon	Complete joint original-model numerical cover

The relevant standard mathematics is also explicit. Bilinear envelopes are attributed to the established global-relaxation literature (McCormick 1976); their four inequalities are derived in Appendix A. An exact product with a selected discrete inventory is an application of known disaggregation, not a generic new linearization theorem. Likewise, continuous declaration of a variable already determined integrally by the remaining integer decisions is a formulation choice whose premises must be proved. The empirical question is whether the resulting method helps this problem.

The project lineage is narrower than a literature genealogy. The current cold P is the repository's canonical compact writer and original objective evaluator (TailoredExact project 2026b). Inspection of the relevant writer, evaluator, startup bridge and representation headers found no verified mapping that identifies one of the public BRP formulations above as its exact original model. Shared routing ideas are insufficient evidence for that identification. H retains the project's inherited decoder and later constructive/exchange components; the paper does not claim to have invented a full genetic algorithm or to have proved H optimal. The code and evidence maps in the appendix and reproducibility notes separate these inherited components from M-B's representation and complete-method comparison.

3. Problem definition and objective interpretation

Let $N = \{1, \ldots, n\}$ contain service stations and 0 be the depot. Each station has capacity C_i, initial inventory b_i, target inventory $D_i > 0$, and weight ω_i. Each vehicle k has capacity Q_k, begins empty and follows at most one depot-closed route. A station may be left unserved or visited once by one vehicle, with an integer nonzero pickup or drop. Simultaneous pickup and drop at the same station, split service and temporary storage operations are excluded. These are physical restrictions of the problem being evaluated, not conclusions about all rebalancing applications.

Write p_{ki} and d_{ki} for pickup and drop quantities. Final station inventory obeys

$$Y_i = b_i + \sum_k (d_{ki} - p_{ki}), \qquad 0 \le Y_i \le C_i. \tag{1}$$

Every route prefix and its return load must lie in $[0, Q_k]$. A vehicle can collect, deliver and collect again; cumulative pickup is therefore not bounded by Q_k. Loaded return is allowed. If $R_k = \sum_i (p_{ki} - d_{ki})$, empty departure and nonnegative return imply

$$\sum_i Y_i + \sum_k R_k = \sum_i b_i. \tag{2}$$

Station inventory can decrease through depot returns. The capacity/initial values serialized for the depot are excluded from the objective and do not supply a preloaded fleet. In particular, a large depot stock entry cannot repair a station shortage in this model.

Travel time is the Euclidean distance between the serialized coordinates divided by 1.5. With unit handling times t_p, t_d, total duration of route k is

$$\text{travel}_k + t_p \sum_i p_{ki} + t_d \left(\sum_i d_{ki} + R_k \right) = \text{travel}_k + (t_p + t_d) \sum_i p_{ki} \le T. \tag{3}$$

The equality follows from load conservation and includes depot unloading. Thus the form used in the writer is not an approximation that ignores return handling. All reported fleets are replayed with their assigned capacities,

ordered operations and complete durations. The parameter T is physical route time; the computational cap is a separate resource.

Set $r_i = Y_i/D_i$, $S = \sum_i r_i$, and $A = \sum_{i<j} |r_i - r_j|$. The objective is

$$G_{\text{true}}(Y) = \begin{cases} A/(nS), & S > 0, \\ 0, & S = 0, \end{cases} \qquad F(Y) = G_{\text{true}}(Y) + \lambda \sum_i \omega_i |r_i - 1|. \tag{4}$$

The denominator depends on the final inventory vector. The convention at $S = 0$ sets dispersion to zero but retains all relative target penalties. For nonnegative ratios, $0 \le G_{\text{true}} \le (n-1)/n$. The target penalty is two-sided: inventories above as well as below target are penalized. We consistently call it relative target deviation, rather than observed unmet demand.

A zero Gini indicates identical ratios, including the stipulated all-zero case. It does not indicate that stations meet their targets. For example, a common ratio .5 has zero dispersion but positive target deviation when weights and lambda are positive. Conversely, under strictly positive lambda and weights, $F = 0$ requires every ratio to equal one. The study's positive-objective instances cannot be interpreted through a zero-dispersion shortcut.

The weights are max-normalized within the constructed inputs rather than normalized to sum one. Therefore $\lambda \sum_i \omega_i$ varies across the panel. A fixed lambda does not imply a fixed aggregate tradeoff between target deviation and Gini as station count or the weight distribution changes. This matters both for interpreting objective magnitudes and for the difficulty of proving optimality. Cross-instance values of F are not a common operational service score.

The static assumption freezes inventories during redistribution; customer rentals and returns do not alter them while trucks operate. Coordinates have a real station-source basis, but stock levels, targets, weights and the centroid depot follow a synthetic, declared construction. We therefore evaluate a physical optimization model on traceable constructed instances, not an end-to-end deployment validated against realized rental outcomes.

4. Formulations and problem-specific strengthening

4.1 Why the inventory representation matters

The cold baseline P-GRB is the canonical compact MILP for the physical problem in Section 3. Binary arcs and assignments, unique station service, direction indicators, MTZ ranks, prefix loads and the complete duration constraint couple routing to final inventory. Appendix A.1-A.3 gives the full numbered formulation in one place. It retains apparently redundant rows actually written by the implementation; it does not substitute a more familiar routing model. The difficulty is not just route selection: integer station stocks enter both the numerator and the variable denominator of ratio Gini.

Write $H_{ij} \ge \pm(r_i - r_j)$, $e_i \ge \pm(r_i - 1)$, $P = \sum_i \omega_i e_i$, and introduce a Gini epigraph g. Exact products $Z_i = gY_i$ give

$$n \sum_i Z_i/D_i \ge \sum_{i<j} H_{ij}, \qquad \min f = g + \lambda P. \tag{5}$$

Cold P uses binary inventory bits and binary-continuous products. M-B selects one integer inventory from the necessary domain $\mathcal{Y}_i$ in each interval $[\alpha, \beta]$:

$$\sum_{y \in \mathcal{Y}_i} s_{iy} = 1, \quad Y_i = \sum_y y s_{iy}, \quad \alpha s_{iy} \le q_{iy} \le \beta s_{iy}, \quad \sum_y q_{iy} = g, \quad Z_i = \sum_y y q_{iy}. \tag{6}$$

At integer selectors one state y=Y_i is active; all other q are zero and the active q equals g, proving $Z_i = gY_i$. This is an application of discrete product disaggregation, not a new general product theorem. Its LP relaxation differs from the bit model. The actual mode retains the original ratio and deviation constraints rather than replacing them with state-based deviation equalities.

M-B declares pickup/drop continuous while retaining integer Y, unique service, binary assignment and binary direction. At an unserved station both quantities are zero. At the sole serving vehicle either d=0 and p=b−Y, or p=0 and d=Y−b. The nonzero quantity is therefore integer. Integer load declarations are retained but are unnecessary for this inference. Multiple serving vehicles, simultaneous pickup/drop or fractional inventory would invalidate it. Independently, one-way nonzero service proves the state/service equalities

$$\sum_k (p_{ki} + d_{ki}) = \sum_y |b_i - y| s_{iy}, \qquad \sum_k z_{ki} + s_{i,b_i} = 1. \tag{7}$$

A missing initial-inventory state means $s_{i,b_i} = 0$. These equalities link movement and visit to the selected state; they are not extra premises of the integrality argument. ENS-C keeps integer p/d and omits these two equalities, while sharing H, the static package and the controller. Neither a representation comparison nor an ideal implication between rows identifies the independent computational contribution of a constituent without its ablation.

4.2 Physical strengthening and the role of integer service

Conservation gives $\sum_i b_i - \sum_k Q_k \le \sum_i Y_i \le \sum_i b_i$, because the fleet's return stock lies between zero and total vehicle capacity. Visit/inventory links fix Y=b when unserved. Summing route durations gives global handling $h \sum_{ki} p_{ki} \le mT$. Single-station reachability intersects the inventory domain with the largest operation any one vehicle can support after paying its necessary depot round trip. These physical bounds hold before any Gini interval or incumbent is chosen.

Two less immediate families encode route support and supply compatibility. For a two- or three-station support W, the smallest explicit depot-cycle travel $\tau(W)$ is the minimum of two or six orders. Metric shortcutting makes it a lower bound on any full tour containing W. If all W are served, their integer nonzero operations and empty departure yield

$$|W| \le \sum_{i \in W} (p_{ki} + d_{ki}) \le P_k + D_k^{\text{drop}} \le 2P_k, \quad P_k \ge \lceil |W|/2 \rceil. \tag{M1}$$

Thus $\tau(W) + h\lceil |W|/2 \rceil > T$ excludes their simultaneous service. Here P_k is **all** pickup on the route; the supplies may be outside W. The alternative conditional row is

$$h \sum_{i \in W} p_{ki} + M_{kW} \sum_{i \in W} z_{ki} \le T - \tau(W) + M_{kW}|W|, \quad M_{kW} = T + \tau(W) + h \max(1, \sum_{i \in W} \min(b_i, Q_k)). \tag{M2}$$

All served cancels the M terms and leaves a necessary travel/handling inequality. With one missing service, its available right side is at least $2T + h \max(1, \sum_W \min(b_i, Q_k))$, enough for every bounded pickup on W. Appendix A.5.4 proves both branches and gives the exact shared 50000-support enumeration. Capacity limits prefixes, never cumulative pickup.

For transfer, label collected units and match every delivered unit to an earlier pickup on its vehicle; nonnegative prefix load makes such matching possible. A source i for delivery j implies the ordered subsequence 0,i,j,0, whose metric travel plus at least one unit of handling is necessary. Therefore

$$d_{kj} \le \sum_{i \ne j:\ \tau_{0i}+\tau_{ij}+\tau_{j0}+h \le T} p_{ki}. \tag{M3}$$

An empty compatible source set forbids delivery; an unvisited source contributes zero; extra stations cannot manufacture stock. Loaded return consists of unmatched pickups and does not break the argument. This row supplies a relaxation-level connection between stock decisions and route time, while the original ordered loads remain indispensable. Appendix A states the emitted numerical guards, indices and all remaining physical rows.

4.3 Gini and cutoff strengthening through a physical embedding

For nonnegative ratios let $A = \sum_{u<v} |r_u - r_v|$. Fix i<j. For every other u, triangle inequality gives $|r_i - r_u| + |r_j - r_u| \ge |r_i - r_j|$. These n−2 contributions, the pair itself and nonnegative omitted pairs prove

$$A \ge (n-1)|r_i - r_j|, \qquad (n-1)|r_i - r_j| \le n\beta S \quad \text{when } G_{true} \le \beta. \tag{M4}$$

The static H-spread row uses $(n-1)H_{ij} \le n\beta S$. Combining it with the original H lower rows implies both signed variable-S centering rows; $S \le \bar{S}$ then implies fixed centering. The ideal containment is useful for understanding, not a reason to remove current rows or claim measured redundancy. The direct interval cap/floor $n\alpha S \le H_\Sigma \le n\beta S$ and these spread rows admit the **tight physical completion** H=absolute differences.

That qualification is essential. Every physical fleet in the stated true-G interval and cutoff has a completion with g=true G, exact e,H, Z=gY and W_SP=SP, preserving F. An arbitrary native completion may inflate g,e,H or W, and need not survive the added rows. Conversely native integer objective is at least decoded physical F. The proof contract is objective-preserving physical projection, not preservation of every original auxiliary assignment as required by an unrestricted user-cut claim. At S=0 all ratios/differences are zero; g=0 realizes the original convention and the penalty remains payable.

From true $F \le c$ and $G \ge \alpha$, nonnegative penalties give $\omega_i |Y_i/D_i - 1| \le (c-\alpha)/\lambda$. Outward-rounded integer intersections and reachability give the domain rectangle, its penalty floor $\underline{P}$ and S bounds. The active writer performs the finite sequence in Appendix A.5.1; a preset field requesting two propagation rounds does not implement an infinite fixed point or even two feedback rounds. Required movement follows immediately from inventory balance on this tightened rectangle.

For positive S, multiplying F≤c yields $A + n\lambda SP \le ncS$. Replacing S in the denominator by its upper bound gives $A + n\bar{S}\lambda P \le n\bar{S}c$. Four standard McCormick inequalities (McCormick 1976), obtained by expanding the four nonnegative rectangle products, relax W_SP=SP and support the lifted row $H_\Sigma + n\lambda W_{SP} \le ncS$. Appendix A.5.7 gives every coefficient, bound and derivation. The original objective cutoff is retained: at S=0 cross multiplication becomes tautological and would otherwise forget the positive penalty. These are cutoff-scoped rows, regardless of a "Global" implementation metadata label.

4.4 Native and project responsibilities

The project supplies the original model, inventory representation, static physical/interval rows, H, complete verified Starts and domain controller. Gurobi supplies native LP/MIP optimization, presolve, cuts and branching. Native monitoring callbacks can terminate on a mathematical bound target; M-B does not implement dynamic separation of these static families or custom native branching. Benders, pricing, dynamic resource cuts, D1, AMF/RC, single-child contraction and timed-quanta research branches are off. Standard triangle, cover, product and McCormick reasoning is not claimed as new general theory. The method specification matrix maps the finite active list to numbered proofs and actual generation points.

5. Complete M-B algorithm and certificate logic

5.1 Startup and maintained information

H is a finite constructive and neighborhood procedure. It constructs a verified fleet by joint insertion; generates 24 shuffled station/separator orders under seed 20260626 and one completed constructive order; fully decodes each; performs first-improvement guided order descent; then alternates strict physical insertion/quantity closure with neutral balanced relocation/equal-net exchange. Appendix B.6-B.11 supplies the encoding, legal amounts, compensation, full compaction, candidate generation, cache policy and actual ties. This active path does not execute genetic generations. H supplies an own upper bound, never a lower certificate. Every compared integrated arm pays for its own H; saved or cross-arm routes are not free incumbents.

The controller stores own verified fleet, U, its true G, incumbent epoch and a coverage registry. Each unit carries interval, qualified bound and source scope, cutoff, parent/children, depth and status. Runtime caches separately bind canonical matrix identity and complete LP evidence to their generation/epoch. Maintaining this distinction prevents a good LP point from becoming a physical incumbent or a child-domain bound from being promoted to its parent.

5.2 Two bound targets and complete decision procedure

After the selected parent's LP return, let its updated bound be b and the other relevant unit bounds be b_j. Requeue first if any $b_j + \epsilon < b$. Otherwise, until the unit's NEXT milestone has been reached, request

$$t_{next} = \min\{b_j : b_j > b + \epsilon\}, \qquad \epsilon = 10^{-7}, \tag{M5}$$

if this set is nonempty. Once the milestone is reached, or no strict higher competitor exists, allow child lookahead. NEXT therefore raises a controlling unit toward the next frontier level before paying for subdivisions. It is not the child target and not a time slice.

For two qualified finite raw child LP bounds ℓ_j, put

$$\Delta = \max(U - b, \max(\epsilon, 10^{-12})), \quad v_j = \mathrm{clip}((\ell_j - b)/\Delta, 0, 1), \quad \eta = \min(v_0, v_1), \quad \mu = (v_0 + v_1)/2. \tag{8}$$

If $b_+ = \min(\ell_0, \ell_1) \le b + \epsilon$, run terminal MIP. If it strictly improves b and ημ plus the explicitly defined floating score tolerance (B2) reaches .08, split. If the gain is positive but score insufficient, retain the parent and run its MIP with target b+. Recompute decisions from the actual return. With an infeasible child, the frozen policy installs both children atomically and marks the proved empty ones; it does not invoke the dormant contraction variant.

The following algorithms specify the complete active flow. `Native` means the request contract in B.4 (actual model identity, type restoration, remaining global deadline, optional validated complete Start, native return and request cleanup). `Qualify` tests its stated identity/finalization/settings/consistency gates; it is not a new oracle. Geometry, tie order, cache tests and all return cases are fully defined in B.1-B.4.

```
Algorithm 1  FrozenMB
Input: instance and physical parameters; frozen method settings; one process deadline
Output: own fleet/U, qualified complete-cover L, unresolved units and evidence status
1  Run H (B.6), admitting only complete physically verified fleets.
   If own verified zero and the inherited zero-evidence policy passes,
       finalize its independent F>=0 certificate and return.
   If no admissible fleet exists, retain U=+infinity and report startup failure;
       RETURN failure without initializing a finite-cutoff proof.
2  Create root L0=[0,min(U,(n-1)/n)], bound 0, Open, epoch 0;
   retain outside-range analytic floor U. For n=1 use root [0,0].
3  While time remains and unresolved relevant coverage remains:
       select a controlling unit with B.2's epsilon-plateau cyclic tie rule;
       if no unit selectable, stop and retain all remaining obligations;
       outcome := ProcessLeaf(unit);
       FinalizeAndUpdateCover(outcome);
       if deadline, engineering failure or terminal interruption: stop.
4  Finish native cleanup and lifecycle outputs; qualify sources and own witness.
   Return the raw complete-cover minimum L without clipping to U;
       distinguish its numerical policy from exact complement accounting (B.4).
   Return own U/L and all unresolved coverage; certify only if the inherited
       numerical gap, source, physical and lifecycle gates all pass.
```

```
Algorithm 2  ProcessLeaf / ChooseAction
Precondition: selected unit is still represented; no pending native call.
1  Validate/rebuild artifact for current epoch; reuse a qualified same-epoch
   parent LP or request its complete LP relaxation.
   On deadline/invalid LP return OPEN_STOP/ERROR, with parent unchanged.
   On qualified infeasible LP return CLOSE(parent,cutoff evidence).
   Merge a finite optimal parent-LP bound on this same domain.
   If b >= current U-epsilon return FATHOM with its bound source.
2  Apply NEXT rule (M5) in its stated priority order:
       noncontrolling -> REQUEUE;
       strict target -> Native(parent MIP,target NEXT), then RETURN_NATIVE;
       otherwise proceed to 3.
3  If depth>=8 or width<=1e-4+1e-12:
       Native(complete parent MIP,no target), then RETURN_NATIVE.
4  Form midpoint children [alpha,mid],[mid,beta], ids parent.0,parent.1.
   Obtain both complete current LP results or reuse B.2's valid cached pair.
   A child starts with max(parent bound,its qualified finite LP bound),
       but retain raw child LP fields for AM. Children are not active leaves.
   On either incomplete/invalid result return OPEN_STOP/ERROR; keep parent.
5  If either child is qualified infeasible: return SPLIT(children,empty flags).
   Otherwise compute (8), (B2), b+=min(raw child bounds):
       b+<=b+epsilon -> Native(complete parent MIP,no target);
       score+score_tolerance>=.08 -> return SPLIT(both children);
       otherwise -> Native(parent MIP,target b+,kind CHILD).
   For a native action return its actually finalized outcome and source identity.
Postcondition: no unresolved domain removed; no target value admitted as proof
   merely because it was requested; no speculative child counted as active.
```

```
Algorithm 3  FinalizeAndUpdateCover
Input: ProcessLeaf outcome, current cover, own witness/epoch/runtime caches
1  For native outcomes require actual return and request cleanup first.
   On failed engineering qualification: stop ERROR; leave unresolved region.
   Admit any finite qualified native bound only on its recorded domain/cutoff.
   Retain the source cutoff for proof accounting in A.6; do not clip the
       raw stored numerical bound (B.4).
2  Replay a candidate fleet and original objective. If own U improves by >1e-9,
       replace own witness, increment epoch and tighten current cutoffs.
       Keep qualified old scoped floors; invalidate retained models/LP/children
       on next access using B.2; never transfer a child-only floor upward.
3  SPLIT: validate parent identity, both child ids/lineage/endpoints, shared
       midpoint, inherited bounds and full union. Install both in one transaction,
       mark parent Replaced, then mark proved infeasible children Empty.
       Discard replaced/empty backend models; failed transaction retains parent.
4  CLOSE/FATHOM or qualified native OPTIMAL/INFEASIBLE: close only this
       cutoff obligation and keep its complement floor and evidence.
   Target reached + termination requested + bound>=target-epsilon:
       set NEXT milestone only for NEXT, then requeue parent (no forced split).
   REQUEUE: retain all state/coverage, return to selection.
   TimeLimit/interrupted before proof/target: retain eligible partial bound,
       keep unit Open, stop common lifecycle. Unknown/error is never Empty.
5  Recompute full-cover minimum; output unresolved units if any remain.
Postcondition: every potential physical improver is still covered by an open
   obligation or a qualified closed/complement floor.
```

Own-U updates and native-bound admission share a source ledger but not a meaning: a feasible fleet can remain valid even if a dependent proof source is later rejected. Start ineligibility is neither infeasibility nor evidence that a low-G region lacks feasible fleets. The next strict target has priority before geometry eligibility; a narrow/deep leaf can still do a NEXT phase, but eventually uses complete MIP rather than deleting its region.

5.3 Why the returned bounds cover the original problem

For a source bound ℓ on an interval model with objective-preserving embeddings of all physical points in $F \le c$, points inside that cutoff have $F \ge \ell$, while points outside have $F > c$. Its unconditional regional floor is therefore min(ℓ,c). Infeasible cutoff models yield floor c, not an infinite original-region bound. This is why old cutoffs remain in the evidence scope after U improves.

For a complete covering family $\mathcal{C}$ with such floors β_I, plus the analytic outside-range floor U,

$$L = \min\{U, \min_{I \in \mathcal{C}} \beta_I\} \tag{9}$$

is a lower bound. Every physical point belongs to at least one covering region or the complement. Both closed midpoint children preserve the parent's union, including their shared endpoint. Max may combine independent valid floors on the same domain; min combines siblings. An unsplit parent or incomplete lookahead remains an obligation. These observations prove coverage inductively, without treating depth limits, milestones, target termination or repeated U as certificates. Appendix B.5 gives both synthetic teaching states and one labelled R123 atomic-cover event.

5.4 Exact mathematics and numerical evidence

The embedding, triangle, support and cover arguments above concern real/exact arithmetic under their explicit physical premises. Actual matrices use binary64 coefficients and Gurobi tolerances; the retained evidence policy

is 10^{-7}. The implemented penalty closure test $\alpha + \lambda \underline{P} \geq c - 10^{-9}$ can close a non-strict cutoff containing a mathematical point at or just below c. It is a **numerical closure**, not proof of strict rational infeasibility. A local $10^{-9} < 10^{-7}$ comparison does not bound global accumulated error. H's internal floating descent likewise does not prove exact rational monotonicity, although its admitted physical neutral moves preserve the exact recomputed objective and improve the entire duration tuple.

Reported original-problem certificates additionally need own verified U, qualified complete coverage, source/return identities and normal lifecycle completion. The sealed nonzero policy requires −1e−7≤U−L≤1e−7 without clipping; this is stronger than the native certificate function's one-sided test. B.4.1 states every admission gate, the no-native clean own-zero branch $U \leq 1e - 12$, and the exact-zero source-reduced fallback. Invalid proof dependencies cannot be repaired by a good incumbent or independent zero floor; source-reduced outcomes remain separate. The single deadline guarantees an operational end, not certification within every window. No solver tolerance or historical certificate field is changed in this revision. The finite sample and nonfactorial design in Section 6 limit empirical attribution; these mathematical details add no new performance evidence.

6. Direct certification evidence on six fixed small problems

6.1 Prospective design and the one-method question

M-B is the sole proposed method throughout this paper. Before generating any Round126 input or seeing its difficulty, we fixed the six problems, two method entries, order, Seed and budgets in Table SC1. Cold P-GRB is the complete original compact model. It receives no H, incumbent, route, bound, Start, certificate, cache or prior solve from M-B. M-B pays for its own unchanged H, VD-P, implicit integer service, A/B, static strengthening, NEXT/AM, full cover and outputs on every problem. ENS-C is a predecessor/representation comparator retained in the historical section; it is not selected per instance or run in this panel. D1 selection is closed, D0 is not a fallback, and no engine switch or fusion is used.

Table SC1. The entire fixed Seed0 panel. Pickup/drop are 60/60 seconds and lambda is .15 in every row. P denotes cold P-GRB. Each method occupies the first position three times.

Problem	Source	n/m	Geometry / stock	Q	Physical T (s)	Production cap (s)	Order
SC-C15-S	CITI	15/2	compact / shortage	[30, 30]	3600	600	P, M-B
SC-B15-S	CABI	15/2	regional / shortage	[20, 40]	5400	600	M-B, P
SC-C15-B	CITI	15/2	regional / balanced	[30, 30]	5400	600	P, M-B
SC-B20-S	CABI	20/2	compact / shortage	[30, 30]	5400	1200	M-B, P
SC-C20-S	CITI	20/2	regional / shortage	[20, 40]	7200	1200	P, M-B
SC-B20-U	CABI	20/2	compact / surplus	[30, 30]	7200	1200	M-B, P

Table SC1b. All six station-stock and weight totals. Depot placeholders are excluded. Weights are maximum-normalized, not sum-normalized.

Problem	sum initial	sum target	sum weights	lambda sum weights
SC-C15-S	157	178	9.599999	1.43999985
SC-B15-S	99	114	8.98	1.347
SC-C15-B	276	276	6.844233	1.02663495
SC-B20-S	191	217	6.98892	1.048338
SC-C20-S	231	263	7.721076	1.1581614
SC-B20-U	149	129	6.270832	0.9406248

All six inputs were generated once from the two already safe frozen station tables. CITI source identity is `9f9aad24e61d661971c24c6f5ec78a69081edf01ffc433563e52a70b2eaca2d0`. CABI source identity is `d4e664c3ecb8c580e95d39541555be9cf592666a515cb854a857fb1d117ff0b8`, with safe-table identity `d982051e5871c575ecdf9df538605401bd98341ae9b3b48112482eff80d42ffa`. The family is `round126-small-certification-v1|source=<source>|source_sha256=<identity>`, and each selection ID is `r126_<problem>_<geometry>_r<replicate>`. The family enters the original anchor hash and the real selection ID enters target/inventory hash material. The adapter is `round126-thin-offline-adapter-v1`.

The original four anchors, compact/regional geometry, tie rules, three-decimal coordinates, target fractions [.44,.56], total correction/sign repair, maximum-normalized six-decimal weights, centroid depot and distances are retained. Only the declared family/ID and table parameters change. Source capacities and coordinates are geographical data; inventories, targets, weights, depot and fleet settings are constructed. No H, LP, MIP or difficulty filter chose or redrew a problem. Station overlaps are recorded. The six problems are not a complete factorial design, IID city samples or six cities.

For the four shortage problems, service stations alone satisfy s=sum D-sum b>0. Empty departure and nonnegative loaded return imply sum Y<=sum b. Thus, with delta=min(lambda*omega_i/D_i)>0,

$$F(Y) \geq \sum_i (\lambda \omega_i / D_i)|Y_i - D_i| \geq \delta \sum_i |Y_i - D_i| \geq \delta s > 0.$$

Table SC2. Independent positive-objective arithmetic. The coefficients are the exact decimal interpretation of the serialized weights and lambda=.15; parsed binary64 values are also recorded. These floors are not injected into the solver or substituted for its reported L. Empty fleets remain legal, so shortage does not mean infeasibility.

Shortage problem	sum b	sum D	s	Exact-decimal positive floor
SC-C15-S	157	178	21	0.111999825
SC-B15-S	99	114	15	0.09
SC-B20-S	191	217	26	0.0864264375
SC-C20-S	231	263	32	0.089256

Balanced and surplus objectives were not presumed zero or positive. A qualified H zero termination is labelled startup-zero and supports termination of the complete method, without credit to a nonzero native proof or interval backend.

6.2 Complete production clocks and certificate admission

The production executable and DLL are the unchanged frozen identities specified in Section 5 and the reproducibility record. Both entries use Gurobi Seed0, Threads1, PresolveAuto and the original remaining parameters, requesting relative/absolute MIP gaps zero. Evidence tolerance remains 1e-7. H retains seed 20260626. The 30-second native reserve, cap-minus6 CLI request, cap-minus2 external hard stop and common absolute deadline

are inherited; preparation is not moved outside the clock. A larger cap can change a path and is not treated as a freely reusable prefix.

T_produce runs from full method start through normal output sealing, including own H, all models/LP/MIP work, static rows, Start, online verification, callbacks, cleanup and required output. T_verify is independent post-seal validation without new optimization or route search. T_total is their sum; cross-arm reconciliation is separately billed group work. Define T_cert=T_produce only when the final certificate passes every own physical, numerical, identity, scope, cover, return and lifecycle gate; otherwise T_cert is null. It is not first incumbent discovery, Gurobi Runtime or the final MIP call's runtime.

Clean nonzero certification uses the inherited signed-gap admission -1e-7<=U-L<=1e-7 alongside all other gates. Raw L is not clipped to U. Clean own-zero and damaged-source exact-zero retain their distinct inherited policies. Same-input cross-arm witnesses are reconciled only after both own productions, using the unchanged 1e-7 scope/consistency rule; they never supply the other arm's starting information. A damaged bound cannot be repaired by a good fleet or another certificate.

The five main categories are BOTH_CERTIFIED, MB_ONLY_CERTIFIED, P_ONLY_CERTIFIED, BOTH_OPEN and UNEVALUABLE. Both-certification speedup is T_cert(P)/T_cert(M-B), so values above one favor M-B. We display every exact observed time difference and ratio; the descriptive materiality threshold is max(30 seconds,.10*T_cert(P)), producing MB_FASTER/P_FASTER/TIME_CLOSE. This threshold is neither statistical significance nor an engineering gate. Open U/L/gap comparisons are supplementary and generate no main WIN. A normal open native return before its nominal cap is reported at its actual duration: neither the cap nor actual open duration is substituted as a hypothetical certification time, exact speedup or unproved speedup lower bound.

6.3 All twelve arms, six pairs and evidence limits

Table SC3. Complete own endpoints. All signed gaps are retained. Full-precision CSV and source/call/model/fleet indices are part of the package.

Problem	Method	Own U	Qualified L	Raw signed U-L	Certificate / evidence
SC-C15-S	P-GRB	0.462806408538	0.462806408538	1.11022302463e-16	yes / CLEAN_NONZERO_CERTIFICATE
SC-C15-S	M-B	0.462806408538	0.462806408538	5.55111512313e-17	yes / CLEAN_NONZERO_CERTIFICATE
SC-B15-S	M-B	0.94700467634	0.94700467634	1.55431223448e-15	yes / CLEAN_NONZERO_CERTIFICATE
SC-B15-S	P-GRB	0.94700467634	0.94700467634	8.881784197e-16	yes / CLEAN_NONZERO_CERTIFICATE
SC-C15-B	P-GRB	0.0067607771599	0.0067607771599	-1.73472347598e-17	yes / CLEAN_NONZERO_CERTIFICATE
SC-C15-B	M-B	0.0067607771599	0.0067607771599	1.56125112838e-17	yes / CLEAN_NONZERO_CERTIFICATE
SC-B20-S	M-B	0.315389121053	0.315389106185	1.48683525492e-08	yes / CLEAN_NONZERO_CERTIFICATE
SC-B20-S	P-GRB	0.340249880289	0.209049610255	0.131200270034	no / RELIABLE_POSITIVE_LB_OPEN
SC-C20-S	P-GRB	0.280579842119	0.161051808386	0.119528033733	no / RELIABLE_POSITIVE_LB_OPEN
SC-C20-S	M-B	0.216430184034	0.207366884943	0.00906329909051	no / RELIABLE_POSITIVE_LB_OPEN
SC-B20-U	M-B	0.173761789119	0.173761789119	4.16333634234e-16	yes / CLEAN_NONZERO_CERTIFICATE
SC-B20-U	P-GRB	0.221769088877	0.0133968443922	0.208372244485	no / RELIABLE_POSITIVE_LB_OPEN

Table SC4. All three clocks and certification costs (seconds). Unknown certification time is shown as unknown, never replaced by the cap.

Problem	Method	Cap	T_produce	T_verify	T_total	T_cert
SC-C15-S	P-GRB	600	464.293373	6.0499092	470.343282	464.293373
SC-C15-S	M-B	600	41.5624639	1.1568301	42.719294	41.5624639
SC-B15-S	M-B	600	6.85265371	0.9079541	7.76060781	6.85265371
SC-B15-S	P-GRB	600	12.9786823	0.3325261	13.3112084	12.9786823
SC-C15-B	P-GRB	600	243.923894	0.3062603	244.230154	243.923894
SC-C15-B	M-B	600	44.0815251	0.9998445	45.0813696	44.0815251
SC-B20-S	M-B	1200	1069.00171	5.1936419	1074.19536	1069.00171
SC-B20-S	P-GRB	1200	1170.67074	23.7145084	1194.38525	unknown
SC-C20-S	P-GRB	1200	1173.28631	1.3285943	1174.6149	unknown
SC-C20-S	M-B	1200	1170.76201	4.9794916	1175.7415	unknown
SC-B20-U	M-B	1200	18.8735701	1.7018773	20.5754474	18.8735701
SC-B20-U	P-GRB	1200	1170.65504	1.7020734	1172.35712	unknown

Table SC5. All six fixed paired outcomes. Positive P-minus-M-B means less M-B production time. Single-sided categories are qualitative censored observations; failures are UNEVALUABLE rather than ordinary censoring.

Problem	Main category	P minus M-B (s)	T_cert P/M-B	Time description	T_total P/M-B
SC-C15-S	BOTH_CERTIFIED	422.730909	11.1709781	MB_FASTER	11.0100902
SC-B15-S	BOTH_CERTIFIED	6.12602859	1.89396442	TIME_CLOSE	1.71522756
SC-C15-B	BOTH_CERTIFIED	199.842369	5.5334722	MB_FASTER	5.41754069
SC-B20-S	MB_ONLY_CERTIFIED	unknown	unknown	censored / no time ratio	unknown
SC-C20-S	BOTH_OPEN	unknown	unknown	censored / no time ratio	unknown
SC-B20-U	MB_ONLY_CERTIFIED	unknown	unknown	censored / no time ratio	unknown

The fixed denominator is six problems and twelve arms. The pairs comprise 3 both-certified, 2 M-B-only, 0 P-only, 1 both-open and 0 unevaluable observations. M-B certifies 5/6 problems, including 5 nonzero and 0 zero objectives (0 startup-zero). Cold P certifies 3/6, including 3 nonzero and 0 zero objectives. A problem certified by both remains one problem. All certified physical values appear in Table SC3 and the summary, rather than being pooled across instances.

The BOTH_CERTIFIED intersection is {SC-C15-S, SC-B15-S, SC-C15-B}, size 3. Its observed speedup median is 5.5334722 and geometric mean 4.89200527. This conditional intersection does not estimate a whole-panel mean. The neighboring single-sided and open counts above remain part of the result; no cap filling, PAR-2 time, best-of-Seed or selected-problem average is used. T_total reverses the observed production-cost ordering on SC-C20-S. Both cost views are retained. On an open pair this is an output-cost observation, with no certification-time ordering or speedup inference; T_produce remains the main certification clock where qualified.

The evidence provides limited direct certification support on SC-C15-S, SC-C15-B, SC-B20-S, SC-B20-U. No problem meets the predeclared opposite material-time or P-only category. This does not turn the remaining problems into wins or establish a population runtime claim.

The preregistered evidence classification is `LIMITED_DIRECT_CERTIFICATION_SUPPORT`, with w=4 and l=0. The execution state is `COMPLETE`. These are bounded evidence labels, not acceptance standards or universal stability tests. The mathematical propositions establish representation, valid embedding and complete proof obligations under their premises; the measured speed/certification counts are empirical statements. This cold-baseline comparison measures the complete method and does not identify the separate causal contribution of cuts, AM or H. All four shortage cases remain visible, including the both-open SC-C20-S.

7. Extended endpoint evidence and the known long-window proof boundary

7.1 Earlier R122 prospective panel and measurement

The earlier R122 panel (TailoredExact project 2026c) was fixed before its six new inputs were generated once. It uses two frozen station sources: CITI with 442 eligible physical stations from 443 original rows, and CABI with 865 stations. Source identities, station selections and overlap records remain in the input manifest. The sources provide geographic station information; inventory landscapes and targets are constructed by the declared recipe.

The recipe retains four-anchor/hash selection and tie rules, target fractions .44-.56, coordinates serialized to three decimal places, weights to six decimals with maximum normalization, a centroid depot, stock-total correction and the declared heterogeneous or homogeneous capacity vectors. No H, LP or MIP screening selected problems by difficulty. There was no outcome-driven redraw, salt change or best-of-replicate selection.

Table 2. Six new Seed0 problems. Stock labels describe the construction. Lambda is .15 and pickup/drop are 60/60 throughout.

Role	Source	n/m	Geometry / inventory	Q	Physical T s	Cap s	sum weights	lambda sum weights
PB-C20-B	CITI	20/2	compact / balanced	[30, 30]	3600	900	7.288265	1.09323975
PB-B20-S	CABI	20/2	regional / shortage	[20, 40]	7200	900	7.284024	1.0926036
PB-C35-U	CITI	35/3	regional / surplus	[20, 25, 30]	10800	2400	11.590113	1.73851695
PB-B35-B	CABI	35/3	compact / balanced	[25, 25, 25]	7200	2400	14.72	2.208
PB-C50-S	CITI	50/4	compact / shortage	[30, 30, 30, 30]	10800	3600	20.216045	3.03240675
PB-B50-U	CABI	50/4	regional / surplus	[20, 25, 35, 40]	18000	3600	22.597221	3.38958315

Each new problem has four complete arms: cold P-GRB, ENS-C, M-B and MB-D1. Two additional fixed blocks repeat PB-B35-B and PB-C50-S with solver Seed1. Thus there are six new problems, eight method/Seed blocks, 32 arms and 48 directed pairs. The repeats are sensitivity observations on known inputs, not two new problems. Within a block all methods share the same physical input and production cap. Serial method order rotates across the planned blocks; the complete schedule is retained with the published design.

All methods use the same frozen executable and Gurobi 13.0.2 DLL, one native thread, one MIP thread, PresolveAuto, the stated Seed, FeasibilityTol/OptimalityTol 10^{-6}, IntFeasTol 10^{-5}, and requested MIPGap/MIPGapAbs zero. The evidence policy remains 10^{-7}. The common 30-second native reserve is part of the one process deadline. Larger caps can change a method's search path, so a prefix of a longer run is not assumed to reproduce a shorter run.

Production time T_{produce} includes all own online startup, models, LP/MIP work, Starts, callbacks, cleanup and sealing. Post-seal verification is T_{verify}; T_{total} is their sum. Cross-arm reconciliation is group cost, and nested native timers are not added again. A verification tail beyond the native reserve is not a producer timeout. We preserve original final output fleets as the primary U endpoints, even where a separately legal historical witness is slightly better.

The machine/software identity and full clocks are documented in Supplement S and the reproducibility notes. The comparison is within matched complete-method windows, rather than a mixture of historical executables, hardware states and caps. The finite nonfactorial design varies source, scale, geometry, stock, Q, T and aggregate weight together. It does not isolate their individual causal effects or estimate a population win probability.

7.2 Comparison and evidence tiers

The historical endpoint rule, retained only for these older tables, is certificate first. A candidate-only qualified certificate wins; a control-only certificate is a severe loss. When both certify, the time-materiality threshold is $\max(30\text{ s}, .10T_{\mathrm{control}})$. Severe certified regression additionally requires twice the control time and at least 120 seconds excess. Actual time ratios are meaningful only for two certified results.

For two open results let $\mathrm{gap} = U - L$, $a_U = \max(.001, .01|U_{\mathrm{control}}|)$, and $a_{\mathrm{gap}} = \max(.001, .10|\mathrm{gap}_{\mathrm{control}}|)$. Material improvement in U or gap without opposing material deterioration is WIN; the reverse is LOSS; opposing changes are MIXED; otherwise TIE. Severe open regression requires both specified U and gap deteriorations, with thresholds $\max(.001, .05|U_{\mathrm{control}}|)$ and $\max(.005, .25|\mathrm{gap}_{\mathrm{control}}|)$. Missing necessary evidence remains UNEVALUABLE. These are experiment-selection/materiality labels, not statistical significance tests.

We distinguish nonzero numerical certification, an independent own-zero certificate, reliable open bounds, source-reduced evidence and missing evidence. The R122 panel has seven nonzero certified arms and 25 open arms; it has no zero certificate, source-reduced arm or missing arm. All seven certificates are at 20 stations. The Seed0 layer contains 24 arms; the Seed1 layer contains eight open arms. PB-B35-B Seed0 cold P uses the explicitly independent lower floor zero, rather than a positive native proof. The tiny positive Seed1 bound is retained.

7.3 All historical outcomes and adverse cases

Table 3. All 32 published R122 arms. N denotes a nonzero numerical certificate and O a reliable open result, including the stated independent floor. U/L and signed gap are role-specific; caps are production windows. Supplement S1 retains full precision and $T_{\mathrm{verify}}, T_{\mathrm{total}}$ for every row. The displayed tiny signed gaps are not clipped.

Role	Seed	Method	Cap s	U	L	Signed gap	Tier	T_produce s
PB-C20-B	0	P-GRB	900	0.2246803118	0.1765783752	0.04810193667	O	872.58902
PB-C20-B	0	ENS-C	900	0.2246803118	0.2246803118	-1.110223025e-16	N	430.55759
PB-C20-B	0	M-B	900	0.2246803118	0.2246803118	1.054711873e-15	N	395.85954
PB-C20-B	0	MB-D1	900	0.2246803118	0.2246803118	2.498001805e-16	N	665.06878
PB-B20-S	0	ENS-C	900	0.8631523206	0.8631523206	-2.220446049e-16	N	4.6224331
PB-B20-S	0	M-B	900	0.8631523206	0.8631523206	-2.220446049e-16	N	2.6319677
PB-B20-S	0	MB-D1	900	0.8631523206	0.8631523206	-2.220446049e-16	N	2.6983887
PB-B20-S	0	P-GRB	900	0.8631523206	0.8631523206	1.110223025e-16	N	116.03278
PB-C35-U	0	M-B	2400	0.1696364022	0.08395185429	0.0856845479	O	2372.5264
PB-C35-U	0	MB-D1	2400	0.1793004366	0.08353399353	0.09576644305	O	2370.6204
PB-C35-U	0	P-GRB	2400	0.1747718876	0.05970290317	0.1150689844	O	2370.6162
PB-C35-U	0	ENS-C	2400	0.1642374878	0.08157539377	0.08266209402	O	2370.5989
PB-B35-B	0	MB-D1	2400	0.2345814099	0.09200798362	0.1425734262	O	2372.4404
PB-B35-B	0	P-GRB	2400	0.3055557782	0	0.3055557782	O	2370.5561
PB-B35-B	0	ENS-C	2400	0.2465044704	0.132287968	0.1142165024	O	2370.5527
PB-B35-B	0	M-B	2400	0.2465044704	0.1276343018	0.1188701686	O	2371.7862
PB-C50-S	0	MB-D1	3600	0.3723417755	0.3568747399	0.01546703556	O	3572.8812
PB-C50-S	0	M-B	3600	0.3723417755	0.3568747399	0.01546703556	O	3570.5953
PB-C50-S	0	ENS-C	3600	0.3723417755	0.3579807999	0.01436097562	O	3571.8954
PB-C50-S	0	P-GRB	3600	0.4259500753	0.3492206958	0.07672937949	O	3583.0202
PB-B50-U	0	P-GRB	3600	1.176955759	0.06837938241	1.108576376	O	3572.991
PB-B50-U	0	MB-D1	3600	0.677298419	0.4712348696	0.2060635494	O	3570.563
PB-B50-U	0	M-B	3600	0.6974735096	0.4553454	0.2421281096	O	3571.8624
PB-B50-U	0	ENS-C	3600	0.8325480152	0.4523603674	0.3801876478	O	3570.5731
PB-B35-B	1	ENS-C	2400	0.2345814099	0.1146410114	0.1199403984	O	2372.9734
PB-B35-B	1	P-GRB	2400	0.2797984344	2.027189664e-18	0.2797984344	O	2370.5117
PB-B35-B	1	MB-D1	2400	0.2345814099	0.138322042	0.09625936783	O	2370.6168
PB-B35-B	1	M-B	2400	0.2465044704	0.0982245518	0.1482799186	O	2370.4896
PB-C50-S	1	M-B	3600	0.3723417755	0.357008334	0.01533344153	O	3572.6028
PB-C50-S	1	ENS-C	3600	0.3723417755	0.358161762	0.01418001348	O	3570.6597
PB-C50-S	1	P-GRB	3600	0.3917805586	0.3494508596	0.04232969893	O	3570.6723
PB-C50-S	1	MB-D1	3600	0.3723417755	0.357008334	0.01533344153	O	3572.5642

Table 4. Directed-pair classifications by denominator. All 48 individual records, including the nine LOSS/MIXED observations, are in Supplement S2.

Seed	Denominator	Pair	WIN	LOSS	MIXED	TIE	UNEVALUABLE
0	6 new problems	MB-D1/P-GRB	5	0	1	0	0
0	6 new problems	M-B/P-GRB	6	0	0	0	0
0	6 new problems	ENS-C/P-GRB	6	0	0	0	0
0	6 new problems	MB-D1/M-B	1	2	1	2	0
0	6 new problems	MB-D1/ENS-C	1	2	1	2	0
0	6 new problems	M-B/ENS-C	1	1	0	4	0
1	2 fixed repeats	MB-D1/P-GRB	2	0	0	0	0
1	2 fixed repeats	M-B/P-GRB	2	0	0	0	0
1	2 fixed repeats	ENS-C/P-GRB	2	0	0	0	0
1	2 fixed repeats	MB-D1/M-B	1	0	0	1	0
1	2 fixed repeats	MB-D1/ENS-C	1	0	0	1	0
1	2 fixed repeats	M-B/ENS-C	0	1	0	1	0

M-B/P and ENS-C/P each produce six WIN on the six new roles. These are the historical complete-endpoint rule results, which combine certification and open-endpoint comparisons. They do not mean that all six problems certified faster, and they are not the direct certification outcome of Section 6. The two fixed Seed1 repeats retain WIN against cold P for each integrated method, but add no independent problems. M-B versus ENS-C has one WIN, one LOSS and four TIE on new roles; over all eight blocks it has one WIN, two LOSS and five TIE. Thus the chosen paper reference does not dominate the unchanged default instance by instance.

On PB-C20-B, all methods have the same final physical value, approximately .2246803118. M-B, ENS-C and D1 certify, while cold P remains open. M-B takes 395.8595 production seconds and ENS-C 430.5576; their difference is below the declared materiality threshold. D1 takes 665.0688 seconds and loses both incremental certified-time comparisons. Cold P's open run supplies no ratio of times to optimality.

PB-B20-S gives the other certification case. All four methods certify approximately .8631523206. Cold P takes 116.0328 production seconds, compared with 2.6320 for M-B, 4.6224 for ENS-C and 2.6984 for D1. This case permits actual certified-time comparisons. The small integrated-method differences fall below the 30-second minimum materiality floor; the panel's TIE labels should not be mistaken for identical measured times.

The V35 cases expose tradeoffs. PB-C35-U is open for every method. M-B has better U and gap than cold P, yet loses to ENS-C because its final U and gap are worse at the material scale. D1/P is MIXED: its weaker physical endpoint accompanies a stronger lower endpoint and smaller gap. D1 loses to both M-B and ENS-C. These outcomes remain even though the complete integrated family improves over cold P.

On PB-B35-B Seed0, D1 has a better U than M-B and ENS-C but a sufficiently weaker lower bound and larger gap that both incremental pairs are MIXED. With Seed1, those two D1 comparisons become WIN, and M-B/ENS-C changes from TIE to LOSS. All remain open. These fixed repeats show meaningful sensitivity without authorizing selection of the better Seed.

PB-B50-U supplies the sole new-role material D1/M-B WIN: D1 improves both physical U and the lower endpoint relative to M-B, and also improves on ENS-C. It nevertheless remains open. PB-C50-S supplies common integrated gains without natural D1 installation at either Seed. Such an unexposed candidate result is still a valid complete-method observation; no extra probe is forced to manufacture mechanism exposure.

D1's five new-role exposures and its two V20 certificates do not satisfy the declared promotion requirement of at least two exposed new roles with an incremental M-B win or added certificate. Only PB-B50-U qualifies. The completed panel therefore retains the MIXED selection conclusion, with M-B as the existing paper reference and ENS-C as project default. D1 keeps research/ablation value. D0 is not a reserve selection, and the discontinued MB-PC development is not revived.

Historical evidence is used only to constrain attribution. Earlier paid-start/backend comparisons distinguish the value of an own feasible startup from backend representation changes; they do not justify assigning an integrated win to an unexposed row. Earlier D1 C2 Seed0 severe loss and later Seed1 gain, and the F2 D0 Seed reversal, remain separately scoped adverse observations. We do not combine their executables or windows with this panel to construct a formal acceleration factor.

The completed five-role historical paid-start layer is MIXED for both integrated methods against P-S, which pays the same complete H before the original compact backend. Some tailored results improve proof completion after matched startup, while other roles favor P-S. Rejected native positive bounds and surviving independent own-zero witnesses keep their separate evidence grades. That layer also spans different measurement policies and cannot yield a pooled exact speedup. The finite attribution note identifies the concrete source reports and their limitations.

7.4 One known positive-objective long-window problem

The separate long-window study (TailoredExact project 2026a) concerns PB-C50-S only, with 50 stations, four capacity-30 vehicles, physical $T = 10800$, solver Seed0, and three freshly paid 18000-second caps in the fixed order P-GRB, M-B, ENS-C. No previous route, bound, tree, basis or cutoff is loaded. The study adds zero independent problem samples and does not include D1.

Its station totals are $\sum b_i = 495$ and $\sum D_i = 562$. Empty departure and nonnegative return imply $\sum Y_i \leq 495$; therefore $\sum_i |Y_i - D_i| \geq 67$. Positive lambda and weights give

$$F \geq 67 \min_i(\lambda\omega_i/D_i) > 0. \tag{10}$$

The prior exact-decimal explanation gives .24814831875, with a tighter inventory-only penalty floor .26064825 already documented. These are weaker than the published production bounds and are not injected into this method. Their purpose is to rule out an independent zero certificate for this role. No new inventory relaxation or route optimization is performed in preparing the paper.

Table 5. One known-problem long-window study. All three endpoints are reliable and open. No row is a time to optimality.

Method	Cap s	U	L	Signed gap	Certified	T_produce	T_verify	T_total
P-GRB	18000	0.37764170855	0.350377933041	0.0272637755094	False	17971.7112989	1.96611939999	17973.6774183
M-B	18000	0.372341775499	0.358011485552	0.0143302899472	False	17970.5725766	40.964555	18011.5371316
ENS-C	18000	0.372341775499	0.358943666168	0.0133981093313	False	17970.8751446	58.2420274	18029.117172

Under the same fixed rule, M-B/P and ENS-C/P are WIN, and M-B/ENS-C is TIE. M-B improves U over cold P by .005299933051160555 and raises L by .00763355251102632; the signed gap decreases by .012933485562186875. ENS-C has the same final U as M-B and a .000932180615873357 higher L. The latter difference is below the pair's gap-materiality threshold. Arithmetic endpoint gains are not ratios of solver speed, and F* remains unknown.

M-B's final-quality fleet appears in the raw process record at 34.328 seconds; its qualified arm-availability upper time is 35.7173270817 seconds. No better own U is observed afterward. This does not establish that U is optimal, nor that all remaining work is pure proof. ENS-C has an early raw startup record at 34.312 seconds, but its clock-origin spread of 1.5845544376 seconds fails the predeclared one-second conversion check. Its four pre-end arm-clock checkpoints remain unknown. The sealed endpoint and separately measured QPC production/verification clocks remain qualified. We neither widen the guard nor compare those raw times as precise cross-method arm times.

The final coverage explains the outstanding proof rather than extrapolating from a plateau.

Table 6. Actual final integrated-method covers. “Empty” excludes improvement under the applicable cutoff; it does not assert unrestricted physical infeasibility.

Method	Unit	True-G lower	True-G upper	Regional floor	State
M-B	L0.1	0.18617088775	0.372341775499	0.372341775499	empty
M-B	L0.0.1	0.0930854438748	0.18617088775	0.372341775499	empty
M-B	L0.0.0.0	0	0.0465427219374	0.358011485552	open
M-B	L0.0.0.1	0.0465427219374	0.0930854438748	0.358827842293	open
ENS-C	L0.1	0.18617088775	0.372341775499	0.372341775499	empty
ENS-C	L0.0.1	0.0930854438748	0.18617088775	0.372341775499	empty
ENS-C	L0.0.0.0	0	0.0465427219374	0.358943666168	open
ENS-C	L0.0.0.1	0.0465427219374	0.0930854438748	0.366384964768	open

The low interval and its adjacent interval are

$$I_0 = [0, .046542721937380925],$$
$$I_1 = [.046542721937380925, .09308544387476185].$$

Both remain unresolved. The final own $G_{\text{true}} = .050119584194518604$ lies in I_1, so that fleet cannot provide an exact-g Start for I_0. M-B makes seven LP and one MIP request; its I_0 NEXT request uses about 17808.127 seconds. ENS-C makes nine LP and three MIP requests, including both interval targets and a subsequent low-interval terminal request. Child LP count and committed split count are different; counts do not establish retained native search trees or an internal cause of difficulty.

A fixed-endpoint counterfactual ledger illustrates why more than one obligation matters. Hold U and all other regional floors fixed. Raising only M-B's I_0 floor to U leaves $U - L_1 = .013513933206048268$; closing only I_1 leaves the global minimum at the old I_0 floor. This follows from the union minimum in (9). It is not a prediction that a different leaf schedule would perform better, and does not justify launching another experiment.

All three normal long-window observations are complete, qualified open results. The finite study records a proof limit, not an execution failure. The longer cap ends here. There is no final certification-time ratio, linear remaining-time prediction or automatic extension until a certificate appears.

8. Discussion, limitations and reproducibility

The direct empirical statement is the exact six-pair certification distribution and conditional both-certified time ratios in Section 6. Its scope is this original model, fixed small problem set and matched complete production budgets. That comparison is meaningful because the physical problem, input and full production costs are common. It does not establish state-of-the-art superiority against all algorithms solving this same physical model. Neighboring published methods differ in target feasibility, visits, depot supply, time treatment or objective. A matched adaptation would require its own justified design and validation.

The method's exact representation also has a precise scope. Implicit quantity integrality follows from the remaining integer decisions and unique one-way service. Its proof would fail under split service even if the same variable names were retained. Inventory-product exactness is an integer embedding property, while the SP estimator is a continuous relaxation. Local native success is usable globally only through cutoff complementation and full coverage. These distinctions identify what must be rederived before extending the model.

Certification is more limited than feasible-solution improvement. On the earlier R122 panel, all nonzero certificates occur at V20 and every V35/V50 outcome is open. Round126 adds the separately reported V15/V20 direct evidence. The known V50 long study keeps two integrated-method improvement obligations. This does not prove that V50 is intrinsically intractable or that no larger instance can certify; other historical cases remain valid in their own scope. It does rule out describing the present evidence as stable certification of all instances up to 50 stations.

H and the exact backend have complementary roles. H rapidly finds an effective physical fleet on the long role; its unchanged U leaves uncertainty about both primal suboptimality and lower-bound shortfall. Native MIP work dominates the recorded large-instance cost, but its internal search/proof allocation is not fully observable. The incompatible low-interval Start is a geometric fact about true Gini, not proof that all difficulty is primal. Inventory/ratio attainability and full fleet-route attainability must be distinguished before proposing an additional mechanism.

The numerical guarantee should be read literally. Qualified bounds and certificates use the frozen floating-point evidence policy, including the explicitly numerical closure guard. Exact coefficient arithmetic and physical replay support specific algebraic and witness checks; they are not an a posteriori rational certificate for every native

optimization operation. The available evidence reveals no new same-domain counterexample, but the policy supplies no global accumulation theorem merely from the relative sizes of two tolerances.

Generalization is constrained by the experimental design. Each six-problem panel uses non-IID constructed problems from two station tables; neither panel represents six independent cities, and the two panels are not pooled as one certification sample. Station overlaps are recorded, and source, size, inventory regime, geometry, fleet and time vary together. The two R122 fixed-input Seed repeats expose sensitivity but cannot estimate a population probability. There is no best-of-Seed endpoint or instance-wise winner dispatcher. The tested lambda value and varying total weights do not establish robustness across tradeoff preferences. Objectives and time scales should therefore be reported per role.

Real station coordinates do not make the constructed inventories operational observations. Targets are not inferred service-level guarantees, the depot is synthetic, and Euclidean travel ignores the road network and traffic. The static model excludes concurrent customer demand and permits only the stated service semantics. Demographic fairness, station access, realized availability, staff workload and revenue would each need their own data and objective interpretation. The title's ratio dispersion should not be read as an empirical equity outcome.

The manuscript preserves unfavorable scientific evidence while moving engineering detail to its supporting files. Supplement S retains all 32 arms and 48 pairs, all three clocks, the three long endpoints and checkpoint unknowns. The original final-fleet endpoint is retained in the Seed1 cold-P case where an earlier legal witness was slightly better; its verification recovery and waiting costs remain in T_{verify}. This distinction prevents changing an endpoint after observing its trajectory.

Reproduction has two levels. The frozen published campaigns retain detailed native source/call/model/return and fleet evidence (TailoredExact project 2026c, 2026a). R124's separate small dependency set supported three complete long-study endpoint replays and all 51 directed-pair classifications. R125 preserved those tables byte for byte and added a compact method-check package: rational inequalities, sampled saved matrix rows, finite decision states and declared H-move replay. Round126 retains those definitions and checks; its new work is the fixed certification panel and the complete updated reading draft. It does not repeat the full historical reconstruction. It does not claim a fresh native replication or revalidation of every historical matrix row. Exact input/software identities and the separate review scope are supplied in the reproducibility notes and review record.

The next step is the one evidence-based priority stated in the final report; this manuscript does not authorize further solving. The paper is organized so a reader can judge whether the physical modeling choice, specific integration and finite evidence are scientifically useful, and whether one additional matched comparison or sensitivity question is essential. The present open proof alone supplies no mandate to resume solver time or invent another numbered component.

9. Conclusion

We have described a complete tailored method for static multi-vehicle redistribution with inventory-to-target ratio dispersion and relative target deviation. The formulation preserves the original routing, service, load and time semantics. Integer inventory states support exact products, and unique one-way service explains why M-B can relax quantity declarations. A complete scoped proof cover explains how returned native information can support an original-problem numerical bound.

M-B is the paper's single proposed method. On the fixed V15/V20 panel it certifies 5/6 problems against cold P-GRB's 3/6; all six paired outcomes and complete certification times are reported, including the time-close and open cases. The bounded scientific judgment is `LIMITED_DIRECT_CERTIFICATION_SUPPORT`. This empirical statement is separate from the ideal representation/coverage propositions and their floating-point implementation policy.

The earlier six R122 endpoint WIN labels are preserved under their historical definition and cannot be read as six faster certifications. The known V50 long window remains open for every method. These capability boundaries do not erase accurately limited small-instance evidence, and smaller gaps do not substitute for optimality proof. ENS-C remains a historical comparison and the unchanged project default; no per-instance dispatch, D1 reopening or automatic follow-up experiment is introduced.

References and supporting material

The thirteen citation entries are in references.bib. Access level, DOI, source URL, checked sections and limits are recorded in literature_evidence.csv and literature_scope.md. The complete mathematical specification is Appendix A. Supplement S is inherited unchanged from R124's published arithmetic derivation; the present finite check script validates the new method examples. Reproducibility notes, research decision, and review actions keep source identity, readiness and actual review scope separate from the scientific results.

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Appendix A. Complete model, scoped strengthening and proof obligations

This appendix is the single maintained model specification for the manuscript. It transcribes the frozen writer, including constraints sometimes redundant for integer solutions. It does not replace the implemented model by a different vehicle-routing formulation. Equations are mathematical real-arithmetic expressions; A.7 states the separate numerical implementation boundary. Source locations refer to production snapshot 355eefd2be3fe5651195681faf7b3a63ee4032f8 (TailoredExact project 2026b).

A.1 Sets, data and variables

Let $N = \{1, \dots, n\}$ be service stations, $N_0 = N \cup \{0\}$, $K = \{1, \dots, m\}$, and $A = \{(i, j) \in N_0^2 : i \neq j\}$. Depot 0 is absent from every inventory, ratio, weight and Gini sum. Vehicle indexing is shifted by one relative to source indices $0, \dots, M-1$. All vehicle-indexed formulas hold for each $k \in K$; all station formulas hold for each $i \in N$, unless otherwise stated.

Station capacity C_i and initial inventory b_i are integers with $0 \leq b_i \leq C_i$; target inventory D_i is a positive integer no larger than C_i. The parser's depot capacity/stock fields do not provide departure supply. Vehicle capacity $Q_k > 0$ is integer. A vehicle starts empty and returns to the same depot, possibly loaded. Travel time $\tau_{ij} = \|v_i - v_j\|_2 / 1.5$ uses the serialized planar coordinates. The route horizon is T; unit pickup/drop times are t_p, t_d, with $h = t_p + t_d$. Penalty multiplier $\lambda \geq 0$ and station weights $\omega_i \geq 0$ are data; the study uses $\lambda = .15$, positive weights and $t_p = t_d = 60$. No inventory floor is imposed by the compatibility field min_ratio.

The binary variables are arc decisions x_{kij}, station service assignments z_{ki}, and pickup-direction indicators a_{ki}. Source calls the last variable mName(k,i); we rename it a to avoid confusing it with the fleet size. Pickup p_{ki}, drop d_{ki}, final inventory Y_i, and post-service load ℓ_{ki} are nonnegative. The rank u_{ki} is continuous. Define $\bar{p}_{ki} = \min(b_i, Q_k)$ and $\bar{d}_{ki} = \min(C_i - b_i, Q_k)$.

Variable	Domain and bound in the active formulation
x_{kij}, z_{ki}, a_{ki}	Binary, for all stated indices; no self arcs
p_{ki}, d_{ki}	Integer for cold P and ENS-C; continuous for M-B; upper bounds $\bar{p}_{ki}, \bar{d}_{ki}$
ℓ_{ki}	Integer in $[0, Q_k]$, retained by all three methods
u_{ki}	Continuous in $[0, n]$
Y_i	Integer in $[\underline{y}_i, \bar{y}_i]$; plain model has $[0, C_i]$
r_i	Continuous in $[0, C_i/D_i]$
e_i	Continuous in $[0, \max(1, C_i/D_i - 1)]$
$H_{ij}, i < j$	Continuous in $[0, C_i/D_i + C_j/D_j]$
g, Z_i	Continuous; $g \in [0, 1]$ in the unsliced writer, $g \in [\alpha, \beta] \subseteq [0, 1]$ for an interval; $Z_i \in [0, C_i]$
f_{kij}	Continuous in $[0, n]$, F0 connectivity flow only
s_{iy}, q_{iy}	Binary inventory selectors and continuous selected-G products, for $y \in \mathcal{Y}_i = \{\underline{y}_i, \dots, \bar{y}_i\}$, VD-P only
$b_{it}^{\text{bit}}, w_{it}$	Binary inventory bits and continuous products, plain P only
W_{SP}	Continuous in $[\max(0, \underline{SP}), \max\{\max(0, \underline{SP}), \bar{S}\bar{P}\}]$, when the aggregate SP estimator is active

The model objective is $f = g + \lambda P$, where $P = \sum_i \omega_i e_i$; $f, S = \sum_i r_i$, and $H_\Sigma = \sum_{i<j} H_{ij}$ are expressions, not new columns. Constants $\underline{y}, \bar{y}, \underline{S}, \bar{S}, \underline{P}, \bar{P}$ are defined in A.5. Bound-only extrema columns r_min,r_max may still be allocated by the writer; with the shared factory their centering inequalities use the projected pair form below and those columns have only their original bounds $[0, \max_i C_i/D_i]$. No additional vehicle-used binary exists: $v_k = \sum_j x_{k0j}$ is an expression.

A.2 Routing, service and physical resources

The complete routing core is

$$\sum_{j\in N} x_{k0j} = \sum_{i\in N} x_{ki0}, \qquad \sum_{j\in N} x_{k0j} \le 1, \tag{A1}$$

$$\sum_{j\in N_0\setminus\{i\}} x_{kji} = \sum_{j\in N_0\setminus\{i\}} x_{kij} = z_{ki}, \qquad \sum_{k} z_{ki} \le 1, \tag{A2}$$

$$z_{ki} \le u_{ki} \le n z_{ki}, \qquad u_{ki} - u_{kj} + n x_{kij} \le n-1 \quad (i,j\in N,\ i\ne j). \tag{A3}$$

These are the writer's original depot degree, station degree and MTZ rows. Any selected station cycle contradicts the sum of (A3) around the cycle. Thus an integer nonempty selection has one depot-connected elementary tour; an unused vehicle has all arcs and assignments zero. A depot-to-depot empty route is represented without a self arc. The MTZ rows remain when F0 also adds the following connectivity flow:

$$0 \le f_{kij} \le n x_{kij} \quad ((i,j)\in A), \qquad \sum_{j\ne i} f_{kji} - \sum_{j\ne i} f_{kij} = z_{ki}, \tag{A4}$$

$$\sum_{j\in N} (f_{k0j} - f_{kj0}) = \sum_{i\in N} z_{ki}.$$

F0 here means the round20-current flow variant, including return-arc columns, with no $f \ge x$ normalization or start-coupling additions. A route visiting q stations admits downstream-count flows $q, q-1, \dots, 1, 0$. These flows establish validity for every original route, independently of MTZ. Cold plain P does not receive this strengthening. The row package also called F0 has additional families in A.5; these two uses of the historical label must not be confused.

All protected service rows are retained:

$$p_{ki} \le \bar{p}_{ki} z_{ki}, \quad d_{ki} \le \bar{d}_{ki} z_{ki}, \quad a_{ki} \le z_{ki}, \quad p_{ki} \le \bar{p}_{ki} a_{ki}, \tag{A5}$$

$$d_{ki} \le \bar{d}_{ki}(z_{ki} - a_{ki}), \quad p_{ki} + d_{ki} \ge z_{ki}, \quad 0 \le \ell_{ki} \le Q_k z_{ki}.$$

The first pair is not silently dropped even where implied by direction rows. Zero directional bounds cause no division. Unique service, a binary direction and the last service inequality forbid split service, simultaneous pickup/drop and zero-operation visits.

Loads are the writer's original two-sided Big-M recurrences:

$$-Q_k(1 - x_{k0i}) \le \ell_{ki} - p_{ki} + d_{ki} \le Q_k(1 - x_{k0i}), \tag{A6}$$

$$-Q_k(1 - x_{kij}) \le \ell_{kj} - \ell_{ki} - p_{kj} + d_{kj} \le Q_k(1 - x_{kij}) \quad (i,j\in N,\ i\ne j), \tag{A7}$$

$$R_k := \sum_i (p_{ki} - d_{ki}) \ge 0, \qquad \sum_{(i,j)\in A} \tau_{ij} x_{kij} + h \sum_i p_{ki} \le T. \tag{A8}$$

The first selected arc fixes departure load to zero. Selected internal arcs propagate each post-service load. A return has no service operation: its load equals the last ℓ, hence is at most Q_k. Summing the recurrence gives R_k as that load, so its upper bound follows without a separate R_k column or explicit upper row. Inactive recurrence rows are valid because their difference is between a feasible before-service load and another feasible post-service load, each in $[0, Q_k]$. Empty vehicles have $p = d = \ell = R = 0$.

In (A8), let $P_k = \sum_i p_{ki}$ and $D_k^{\text{drop}} = \sum_i d_{ki}$. Since $R_k = P_k - D_k^{\text{drop}}$, original handling time is

$$t_p P_k + t_d(D_k^{\text{drop}} + R_k) = (t_p + t_d)P_k. \tag{A9}$$

Return unloading is paid. Capacity Q_k limits every load, not cumulative pickup P_k, which may exceed Q_k after intermediate drops. There is one physical time limit per vehicle and a separate computational deadline.

Inventories and objective auxiliaries satisfy

$$Y_i + \sum_k (p_{ki} - d_{ki}) = b_i, \quad r_i = Y_i/D_i, \quad e_i \geq r_i - 1, \quad e_i \geq 1 - r_i, \tag{A10}$$

$$H_{ij} \geq r_i - r_j, \quad H_{ij} \geq r_j - r_i \quad (i < j). \tag{A11}$$

Consequently $\sum_i Y_i + \sum_k R_k = \sum_i b_i$. Loaded returns permit a decrease of station inventory; no external depot inventory can increase it.

A.3 Original and disaggregated Gini objective

The physical objective is

$$F(Y) = G_{\text{true}}(Y) + \lambda \sum_i \omega_i |Y_i/D_i - 1|, \quad G_{\text{true}} = \begin{cases} \dfrac{\sum_{i<j} |r_i - r_j|}{n \sum_i r_i}, & S > 0, \\ 0, & S = 0. \end{cases} \tag{A12}$$

The native epigraph is

$$n \sum_i Z_i/D_i \geq H_\Sigma, \qquad \min f = g + \lambda \sum_i \omega_i e_i. \tag{A13}$$

Cold P uses inventory bits. With $B_i = \max(1, \lceil \log_2(C_i + 1) \rceil)$, for $t = 0, \ldots, B_i - 1$,

$$Y_i = \sum_t 2^t b_{it}^{\text{bit}}, \quad Z_i = \sum_t 2^t w_{it}, \quad 0 \leq w_{it}, \quad w_{it} \leq g, \quad w_{it} \leq b_{it}^{\text{bit}}, \quad w_{it} \geq g + b_{it}^{\text{bit}} - 1. \tag{A14}$$

The capacity bound removes unused bit encodings. These products are exact at binary bits. ENS-C and M-B instead use the finite value-disaggregated perspective block VD-P:

$$\sum_{y \in \mathcal{Y}_i} s_{iy} = 1, \quad Y_i = \sum_y y s_{iy}, \quad \alpha s_{iy} \leq q_{iy} \leq \beta s_{iy}, \quad \sum_y q_{iy} = g, \quad Z_i = \sum_y y q_{iy}. \tag{A15}$$

The active mode is VD-P, not VD-J: equalities reconstructing r_i or e_i from selectors are not substituted for (A10). The shared factory's bit-product McCormick rows are skipped for this block because those bit columns are absent. Products (A15) already use interval bounds. M-B additionally writes the original A/B rows

$$\sum_k (p_{ki} + d_{ki}) = \sum_{y \in \mathcal{Y}_i} |b_i - y| s_{iy}, \qquad \sum_k z_{ki} + s_{i,b_i} = 1. \tag{A16}$$

A missing initial state means $s_{i,b_i} = 0$, not a new selector. The independent ENS-Q option is false. The mode-linked direction equality of another variant is absent; M-B retains its original binary a_{ki}. ENS-C uses integer p/d and omits (A16). These are the material representation differences; the complete H and controller are shared.

A.4 Integrality, embedding and fairness propositions

Lemma A1 (implicit service integrality). With integer b_i, Y_i, (A2), (A5), and binary assignments/directions, M-B's continuous quantities are integer.

Proof. If $z = 0$, its quantities and direction are zero. At the only assigned vehicle, direction 1 forces $d = 0$, so $p = b_i - Y_i$ is integer. Direction 0 forces $p = 0$, so $d = Y_i - b_i$ is integer. No other vehicle contributes. Retained integer loads are not needed for this argument. Removing unique service, one-way direction or integer inventory invalidates the inference. The statement concerns integer feasible solutions, not equality of LP convex hulls. □

Validity of A/B. For an integer original service, total movement at a station equals $|b_i - Y_i|$, which its selected state reproduces. An assigned service is nonzero and one-way, so $Y_i \neq b_i$ exactly when the station is served. This proves both (A16); neither is an extra premise of Lemma A1.

Lemma A2 (physical objective embedding). Assume inventory domains and rows preserve all physical points assigned to a true-G interval $[\alpha, \beta]$ and cutoff $F \leq c$. Every such fleet has an extension satisfying the applicable original or VD-P groups, with $g = G_{\text{true}}$ and $f = F$, up to permutation of identical vehicle labels if symmetry rows are present.

Proof. Use the physical arcs, service, load prefixes, tour ranks and downstream flows. Select its integer inventory state, put $q_{iy} = g$ at the selected state and zero at others; (A15) yields $Z_i = gY_i$. Set $e_i = |r_i - 1|$ and $H_{ij} = |r_i - r_j|$. Then (A13) holds, with equality for $S > 0$. At $S = 0$ all inventories, ratios and differences are zero and $g = 0$ realizes the stipulated convention. Bounds and valid strengthening are satisfied by assumption. Conversely an integer product completion has $Z_i = gY_i$, hence $g \geq G_{\text{true}}$ for $S > 0$ and $f \geq F$; at zero sum, $g \geq 0$ suffices. □

Arbitrary native g may exceed physical Gini. A fleet with $G_{\text{true}} < \alpha$ can have an inflated-g auxiliary completion. The proof assigns physical points by their true Gini and requires their objective-preserving extension; it does not classify every completion by its auxiliary g. An inventory vector without compatible routes is not an upper-bound witness.

For $n > 1$, triangle inequalities give $A := \sum_{i<j} |r_i - r_j| \leq (n-1)S$, so $G_{\text{true}} \leq (n-1)/n$. For $S > 0$, $G_{\text{true}} = 0$ means every ratio is equal; at $S = 0$ all are zero by nonnegativity. A common ratio below one is a common proportional shortage, not target attainment. If $\lambda > 0$ and all $\omega_i > 0$, $F = 0$ if and only if every $Y_i = D_i$. These are dispersion and target-deviation statements, with no implied demographic/service/revenue interpretation.

A.5 Enabled strengthening: construction, proofs and scope

The finite list below describes the actual M-B path, not every option in the repository. All rows are assembled **statically when a canonical interval model is written**, before its LP or MIP request. A new cutoff epoch rewrites the relevant model; it does not separate these rows at fractional native nodes. The writer supplies physical rows, and the shared interval factory supplies the interval rows. Families registered by the factory are not all rows emitted by it. In particular, the factory's bit-product rows are filtered out for VD-P. The original objective and physical constraints remain present.

A.5.1 Construction of the inventory rectangle (D1)

Let the current true-G interval be $I = [\alpha, \beta]$, the independently verified own incumbent be U_e, and the cutoff be $c = U_e$. **The active controller sets incumbent_epsilon to zero**, so this is a non-strict objective cutoff; a generic writer can subtract an epsilon, but that is not this frozen request. Start with $l_i = 0, u_i = C_i$. With $\lambda > 10^{-12}$, finite positive U_e, and $B = (c - \alpha)/\lambda \geq -10^{-10}$, for each $\omega_i > 10^{-12}$ intersect with

$$l_i \leftarrow \max\{l_i, \lceil D_i(1-\max(0,B/\omega_i)) - 10^{-9}\rceil\}, \quad u_i \leftarrow \min\{u_i, \lfloor D_i(1+\max(0,B/\omega_i)) + 10^{-9}\rfloor\}. \tag{A19}$$

This is followed by one intersection with physical movement reachability. Write $t_i = \tau_{0i} + \tau_{i0}$ and

$$\begin{aligned} M_{ki} &= \begin{cases} \min\{Q_k, \lfloor (T-t_i)/h + 10^{-9}\rfloor\}, & T + 10^{-9} \geq t_i,\ h > 10^{-12}, \\ Q_k, & T + 10^{-9} \geq t_i,\ h \leq 10^{-12}, \\ 0, & \text{otherwise.} \end{cases} \\ l_i &\leftarrow \max\{l_i, b_i - \max(0, \max_k \min(b_i, Q_k, M_{ki}))\}, \\ u_i &\leftarrow \min\{u_i, b_i + \max(0, \max_k \min(C_i - b_i, Q_k, M_{ki}))\}. \end{aligned} \tag{A20}$$

The outer max with zero transcribes the actual zero-initialized reachability accumulators, including a slightly negative quotient admitted by a floating travel guard. The mathematical version omits the floating guards, uses $h = 0$ for the second case, and clips movement below at zero. Positive published $h = 120$ avoids the small-positive-h threshold ambiguity. If an interval becomes empty, the code flags domain infeasibility, resets that station's storage bounds to $[0, C_i]$, and emits an impossible row. The reset prevents malformed column ranges; it does not reinstate the excluded domain as feasible.

Proposition A5 (budget and reachability). Every physical point in $I, F \leq c$ satisfies the ideal versions of (A19)-(A20). Indeed $F = G + \lambda P$, $G \geq \alpha$, and nonnegative weighted deviations imply $\omega_i |Y_i/D_i - 1| \leq (c-\alpha)/\lambda$. Integer Y_i gives the rounded bounds. A visited station requires a depot tour with travel at least t_i. Its pickup is at most Q_k, b_i, and its drop is at most $Q_k, C_i - b_i$; because $D_k^{\text{drop}} \leq P_k$, either operation is at most cumulative pickup, which has handling cost $hP_k \leq T - t_i$. Maximize over its unique possible vehicle. These are necessary, not sufficient, attainability conditions: a delivery still needs a source. Zero weight or zero lambda supplies no individual penalty bound and is skipped. □

Actual order and pass count. The writer first computes budget bounds, intersects reachability, and derives interval extrema; the shared factory recomputes this same sequence from $[0, C_i]$ and replaces the local arrays. It does not feed the first result into a second tightening pass. There is no propagation loop in this enabled path and no iterative fixed point. The preset advertises `iterative` and requested rounds 2, but the legacy completion field is capped at 1. Thus the operative definition is one budget intersection plus one reachability intersection, independently recomputed by the factory. R124's "original two passes" description was incorrect; production is unchanged.

Set $\underline{y}_i = l_i, \bar{y}_i = u_i, \mathcal{Y}_i = \{l_i, \ldots, u_i\}$, and

$$\begin{aligned} \underline{r}_i &= l_i/D_i, \quad \bar{r}_i = u_i/D_i, \\ \underline{S} &= \sum_i \underline{r}_i, \quad \bar{S} = \sum_i \bar{r}_i, \\ \underline{e}_i &= \operatorname{dist}(1, [\underline{r}_i, \bar{r}_i]), \quad \bar{e}_i = \max\{|\underline{r}_i - 1|, |\bar{r}_i - 1|\}, \\ \underline{P} &= \sum_i \omega_i \underline{e}_i, \quad \bar{P} = \sum_i \omega_i \bar{e}_i. \end{aligned} \tag{A21}$$

The last extrema bound the **tight physical completion** of e. They do not assert that every nonminimal native epigraph e_i is at most $\bar{e}_i$. There is no stock-weighted knapsack or optimization in the denominator calculation: these are sums over the rectangle, with no extra conservation-driven tightening.

A.5.2 Physical conservation and activation (P1-P4)

The following rows hold on the full physical domain, independently of (I,c):

$$\sum_i b_i - \sum_k Q_k \le \sum_i Y_i \le \sum_i b_i; \quad -b_i \sum_k z_{ki} \le Y_i - b_i \le (C_i - b_i) \sum_k z_{ki} \quad (i \in N); \tag{A22}$$

$$h \sum_{k,i} p_{ki} \le mT; \qquad z_{ki} \le 0 \quad \text{if } t_i + h > T + 10^{-9}. \tag{A23}$$

Proof. Each return load lies in $[0, Q_k]$; insert this into conservation (A10). If no vehicle serves i, inventory balance fixes $Y_i = b_i$; otherwise unique assignment makes the visit sum one and the two activation bounds reduce to $0 \le Y_i \le C_i$. Sum (A8) and drop nonnegative travel for global handling. Any nonzero integer service has at least one pickup somewhere on the vehicle: for a drop use empty departure and load nonnegativity. Consequently a tour through i takes at least $t_i + h$. The physical meaning of the historical `transfer_subset_capacity` flag here is this singleton visit exclusion, not a separate general subset-flow formulation. All vehicles and stations are enumerated in increasing index order. These inequalities are implied for integer physical points, but can strengthen the relaxation. The global-handling impossible-row diagnostic based on accumulated required pickup is inactive in the shared-factory path because that legacy accumulator is not populated there. □

A.5.3 Ordered supply compatibility (P5)

For every vehicle k and receiving station j, define

$$\mathcal{C}_{kj} = \{i \in N \setminus \{j\} : \tau_{0i} + \tau_{ij} + \tau_{j0} + h \le T + 10^{-9}\}. \qquad d_{kj} \le \sum_{i \in \mathcal{C}_{kj}} p_{ki}. \tag{A24}$$

All stations passing this test enter the set, even those with zero stock; their pickup column is then fixed at zero. Common T, h, τ make the sets identical across vehicles, but their pickup bounds still use Q_k. No demand assumption or target sign is used.

Proposition A6. (A24) is necessary for a physical fleet under metric travel and empty departure. Label the integer units collected by a vehicle. Scan its route in order, assigning each delivered unit to one previously collected, not yet delivered unit. Every prefix load is nonnegative, so this matching always exists. If a unit received at j came from i, the route includes the ordered subsequence (0,i,j,0). Repeated triangle inequalities give travel at least $\tau_{0i} + \tau_{ij} + \tau_{j0}$, and its at least one collected unit entails $hP_k \ge h$. Hence $i \in \mathcal{C}_{kj}$. Counting matched units at j proves the row; a pickup unit can be delivered only once. Intermediate stations can add travel or receive other units but cannot create supply. Unmatched collected units may return to depot. An unvisited source has $p_{ki} = 0$; an empty compatibility set forces $d_{kj} = 0$. Preloading, cross-vehicle handoffs, or nonmetric shortcut assumptions would break this proof and are excluded. □

The row is a coarse aggregate necessity, not a claim that all compatible pickups can simultaneously reach j. It neither imposes $P_k \le Q_k$ nor substitutes for prefix loads.

A.5.4 Two- and three-station support (P6)

For $W = \{a, b\}$,

$$\tau(W) = \min\{\tau_{0a} + \tau_{ab} + \tau_{b0}, \tau_{0b} + \tau_{ba} + \tau_{a0}\}.$$

For $W = \{a, b, c\}$, take the minimum of the six expressions

$$\tau_{0\pi_1} + \tau_{\pi_1\pi_2} + \tau_{\pi_2\pi_3} + \tau_{\pi_3 0}, \quad \pi \text{ a permutation of } (a, b, c). \tag{A25}$$

These are explicit finite formulas, not new shortest-route optimization calls. Metric shortcutting of a complete tour yields one such order, so its travel is at least $\tau(W)$.

Lemma A7 (integer handling). If all stations in W are served by k, their nonzero integer operations give

$$|W| \le \sum_{i \in W} (p_{ki} + d_{ki}) \le P_k + D_k^{\text{drop}} \le 2P_k, \quad P_k \ge \lceil |W|/2 \rceil. \tag{A26}$$

Here $P_k = \sum_{i \in N} p_{ki}$, not $\sum_{i \in W} p_{ki}$: supplies may lie outside W. Integer P_k gives the ceiling; the LP relaxation alone cannot justify it. Capacity does not bound cumulative pickup. Therefore the writer emits exactly one of the following rows for each selected support:

$$\begin{cases} \sum_{i \in W} z_{ki} \le |W| - 1, & \tau(W) + h\lceil |W|/2 \rceil > T + 10^{-9}, \\ h \sum_{i \in W} p_{ki} + M_{kW} \sum_{i \in W} z_{ki} \le T - \tau(W) + M_{kW}|W|, & \text{otherwise}, \end{cases}$$
$$M_{kW} = T + \tau(W) + h \max\{1, \sum_{i \in W} \min(b_i, Q_k)\}. \tag{A27}$$

Proposition A8 (both branches). The cover branch follows from (A8), (A25), (A26). In the conditional branch, if every station is served, cancel $M_{kW}|W|$. Since full travel is at least $\tau(W)$ and $\sum_W p \le P_k$, the remaining inequality is necessary. If at least one is unserved, put $q = |W| - \sum_W z \ge 1$, $B_W = \sum_W \min(b_i, Q_k)$. The right side after moving the visit terms is $T - \tau(W) + qM_{kW} \ge 2T + h \max(1, B_W) \ge hB_W \ge h \sum_W p$. This proves deactivation for all nonnegative T, h, including $B_W = 0$, empty vehicle and $h = 0$. It does not rely on an invalid cumulative-capacity bound. □

Selection is finite and ordered. For k increasing, enumerate every increasing pair $a < b$, then every increasing triple $a < b < c$. One shared counter starts at zero outside the vehicle loop. The helper returns without adding a row once the counter is 50000; it increments once before choosing cover versus conditional. Outer loops continue (and can still evaluate cycle formulas), but no more rows are added. This is not "stop after 50000 violated covers", not 50000 per vehicle, and not pairs interleaved with triples. At n=50 each full vehicle contributes 1225+19600=20825 supports; the third vehicle receives only the prefix needed to reach the shared limit. Row validity is independent of this selection policy. The dormant exhaustive-subset and rank3 variants have different coefficients/order and are excluded.

A.5.5 Dispersion, spread and centering (G1-G4)

For every pair $i < j$, let $d_{ij} = |r_i - r_j|$. For each $u \notin \{i, j\}$, triangle inequality gives $d_{iu} + d_{ju} \ge d_{ij}$. These pairs are distinct in the unordered numerator; the omitted pairs among the other stations are nonnegative. Hence

$$A = d_{ij} + \sum_{u \ne i,j} (d_{iu} + d_{ju}) + \sum_{u<v;\ u,v \ne i,j} d_{uv} \ge [1 + (n-2)]d_{ij} = (n-1)d_{ij}. \tag{A28}$$

For $n = 2$ the other sums are empty and equality holds. For $n = 1$, spread and centering are skipped. For true $G \in [\alpha, \beta]$ and $S > 0$, $n\alpha S \le A \le n\beta S$. At $S = 0$, all ratios and A are zero; only the interval containing true $G = 0$ needs that physical embedding. With **tight** $H_{ij} = d_{ij}$, the emitted rows are

$$n\alpha S \le H_\Sigma \le n\beta S; \qquad (n-1)H_{ij} \le n\beta S \quad (i<j). \tag{A29}$$

This proves existence of an objective-preserving completion. It does **not** prove that an arbitrary inflated native H_{ij}, even at integer physical variables, satisfies the new row. For example $r=(1,1), g=.5, H_{12}=1$ satisfies the unsliced original epigraph $2gS \ge H$; the true-G-zero cap excludes this completion while preserving $H=0, g=0$. Accordingly these static rows are not advertised as callback user cuts preserving all original native integer completions.

Using the retained $H_{ij} \ge \pm(r_i - r_j)$ in the pairwise row gives the two variable-S rows, and $S \le \bar{S}$ then gives the fixed rows:

$$(n-1)(r_i-r_j) \le n\beta S, \quad (n-1)(r_j-r_i) \le n\beta S; \quad \pm(r_i-r_j) \le \frac{n\beta\bar{S}}{n-1}. \tag{A30}$$

The fixed form is the projection of a common min/max ratio band. The allocated extrema columns do not receive that extended formulation in the shared path. All increasing pairs, forward then reverse, are generated. The ideal implication chain is (A29)+absolute-value lower rows → variable-S → fixed-S. Redundancy in ideal arithmetic is not an ablation and authorizes neither row deletion nor an individual speedup claim. The cap/floor, H-spread and signed-difference rows constrain different auxiliary completions even though they preserve the same tight physical embedding.

A.5.6 Required movement and penalty closure (C1-C3)

For each station, emit only the applicable bound consequence:

$$\sum_k (d_{ki} - p_{ki}) \ge l_i - b_i \quad (l_i > b_i), \qquad \sum_k (p_{ki} - d_{ki}) \ge b_i - u_i \quad (u_i < b_i). \tag{A31}$$

These follow by substituting inventory balance in $Y_i \ge l_i$ or $Y_i \le u_i$. Their scope inherits the domain: a cutoff-derived l_i is not an unrestricted physical bound. If $b_i \notin \mathcal{Y}_i$, A/B forces service, consistently with required movement. If $\mathcal{Y}_i = \{b_i\}$, A/B forces no service. A singleton zero domain and zero operation bounds require no divisions.

For all active finite positive-cutoff requests write

$$g + \lambda P \le c, \qquad P \ge \underline{P}, \qquad e_i \le \max(0, B/\omega_i) \ (\omega_i > 10^{-12}). \tag{A32}$$

The last row uses the same budget activation tests as A19. The floor follows station by station from the distance to 1 in (A21); the original e upper bound is retained and the budget row intersects it. $\alpha + \lambda\underline{P} > c$ proves ideal cutoff infeasibility. The implementation instead tests $\alpha + \lambda\underline{P} \ge c - 10^{-9}$, writes an impossible row, and uses its numerical closure policy. In particular equality to c is not strict infeasibility of the non-strict cutoff set. A.7 explicitly limits this claim. The factory's "Global" metadata on the incumbent row means the entire current model, not all physical objectives at every cutoff.

A.5.7 Scalar objective estimate and SP lifting (C4-C6)

For $S > 0$, multiplying the true cutoff inequality by nS gives

$$A + n\lambda SP \le ncS. \tag{A33}$$

Since $A \ge 0$, $\bar{S} \ge S$ gives $A/(n\bar{S}) \le A/(nS)$, so $A + n\bar{S}\lambda P \le n\bar{S}c$ when $\bar{S} > 0$. Equivalently A33 implies $A \le nS(c-\lambda P) \le n\bar{S}(c-\lambda P)$, because the original cutoff and nonnegative G imply $c - \lambda P \ge 0$. The emitted scalar row is

$$H_\Sigma + n\bar{S}\lambda \sum_i \omega_i e_i \le n\bar{S}c \quad (\bar{S} > 10^{-12}). \tag{A34}$$

The bound on S is the rectangular upper estimate (A21), not a new optimum of S. Tight H/e complete every physical point. If $S = 0$, then $A = 0$, $P = \sum_i \omega_i$, and the original condition is $\lambda P \le c$. Multiplication alone makes A33 tautological and cannot enforce that condition. Thus the original objective cutoff (A32) remains essential. A34 also holds directly at S=0 if that cutoff holds.

Introduce $W = W_{SP}$ and take $W = SP$ on physical completions. The four nonnegative products on the rectangle are $(S-\underline{S})(P-\underline{P})$, $(\bar{S}-S)(\bar{P}-P)$, $(\bar{S}-S)(P-\underline{P})$, and $(S-\underline{S})(\bar{P}-P)$. Expanding them proves respectively

$$\begin{aligned} W &\ge \underline{S}P + \underline{P}S - \underline{SP}, \\ W &\ge \bar{S}P + \bar{P}S - \bar{S}\bar{P}, \\ W &\le \bar{S}P + \underline{P}S - \bar{S}\underline{P}, \\ W &\le \underline{S}P + \bar{P}S - \underline{S}\bar{P}. \end{aligned} \tag{A17}$$

These are the standard McCormick envelopes (McCormick 1976), derived here without attributing an unread theorem number. They are relaxations, not an exact continuous-product identity. The writer bounds W by $w_l = \max(0, \underline{SP})$, $w_u = \max(w_l, \bar{S}\bar{P})$, and emits

$$H_\Sigma + n\lambda W - nc \sum_i r_i \le 0. \tag{A35}$$

Every coefficient is explicit: H coefficients 1; r coefficients $-nc$; W coefficient $n\lambda$; e coefficients in each McCormick row are the corresponding S coefficient times ω_i. The active objective remains $g + \lambda \sum_i \omega_i e_i$ with **ObjCon=0**; W is absent from the objective. Equal S or P bounds cause valid degenerate envelopes. Zero S forces the physical W to zero but does not remove the original penalty. These rows preserve a joint tight completion of g,e,H,W and must not be claimed to preserve all loose auxiliary assignments. No GS or disaggregated-SP lift is enabled.

A.5.8 Symmetry, skipped rows and dependency order

With the active symmetry option, if **all** capacities are equal and $m > 1$, the writer adds $\sum_i z_{ki} \ge \sum_i z_{k+1,i}$ for adjacent vehicles. Common T, travel and handling make those vehicles interchangeable; relabel routes by nonincreasing visits to obtain an objective-preserving representative. Heterogeneous capacities skip the whole writer family. Start normalization may sort within equal-capacity classes, which is harmless relabelling but does not add mixed-fleet symmetry rows. A physically feasible labelled route need not itself satisfy the symmetry rows before relabelling.

Bit-only interval envelopes are $\alpha b \le w \le \beta b$, $g - \beta(1-b) \le w \le g - \alpha(1-b)$. They follow by the cases b=0,1; they are **not emitted in active VD-P**, whose q block already embeds its interval bounds. Dormant

receiver-source covers, Benders, pricing, D1, exhaustive subset duration, dynamic resource cuts, custom branch priorities, S buckets, GS, AMF/RC, and old timed quanta are not part of this specification. Monitoring/target callbacks do not turn the static strengthening into dynamic separation.

The proof dependency is: physical core → implicit integer service and conservation → reachability/transfer/support; budget plus reachability → rectangle → movement/penalty/S bounds; triangle lemma plus true-G membership → Gini rows; rectangle plus original cutoff → SP/scalar rows. A/B use unique nonzero one-way service independently of the implicit-integer lemma. All can be satisfied by the same physical completion, which avoids circularly assuming the validity of a tightened domain to prove its own construction. Analytic validity and actual partial support selection are separate from computational contribution; no family-level performance attribution is made without an ablation.

A.6 Cutoff and coverage proof

Lemma A3 (cutoff complement). Suppose a returned lower bound ℓ is qualified for a model containing objective-preserving embeddings of every physical point with $G_{\text{true}} \in I$ and $F \leq c$. Then $\min(\ell, c)$ is an unconditional regional lower bound.

Proof. A point in the cutoff set has $F = f \geq \ell$. A point outside has $F > c$. The minimum covers both. If the cutoff model is infeasible, only that cutoff set is empty, so c, rather than $+\infty$, is the original-region floor. □

Lemma A4 (complete cover). Let every possible improving true-G value belong to at least one registered unit I, with a qualified unconditional floor β_I. Include the analytic complement, whose applicable floor is U. Then

$$L = \min\{U, \min_{I \in \mathcal{C}} \beta_I\} \tag{A18}$$

is a lower bound on the original optimum.

Proof. Any feasible point is in the complement or at least one covering region. Its corresponding floor bounds its objective. Taking the minimum is safe for the union. Initially $F \geq G_{\text{true}}$ allows $[0, \min((n-1)/n, U)]$ as the improving Gini range. Replacing $[\alpha, \beta]$ by both closed children $[\alpha, \mu]$, $[\mu, \beta]$ preserves coverage, including the shared endpoint. Replacement commits both children atomically with their evidence. An unsplit parent, unresolved leaf, or qualified closed leaf remains represented. Induction over updates proves the invariant. □

Bounds on the same physical domain can be combined by maximum; the union of sibling domains requires minimum. Transfer to a narrower domain requires containing scope and a handled cutoff complement, and does not construct that domain's feasible Start. Proof identities include actual model SHA, input, call, epoch, domain, cutoff and return. A local terminal status cannot certify the whole original problem. Depth/width/scheduling limits can select a request or leave an obligation open; they cannot delete it.

For a compatible own fleet, the mapped complete Start includes unused vehicles and all auxiliaries. Setting $g = G_{\text{true}}$ outside a leaf is not a valid exact-g Start. Submission, API acceptance and an improved native incumbent are different observations. No optimality proof follows from H, repeated physical U or Start acceptance.

A.7 Numerical implementation boundary

Lemmas A1-A4 concern ideal exact arithmetic and valid mathematical domains. Stored matrices use binary64 arithmetic and Gurobi 13.0.2. Native FeasibilityTol and OptimalityTol are 10^{-6}, IntFeasTol 10^{-5}, requested MIPGap/MIPGapAbs zero; retained evidence closure uses 10^{-7}. A qualified result is a numerical certificate under this policy, not a rational exact proof.

In particular, the implemented guard $\alpha + \lambda \underline{P} \geq c - 10^{-9}$ can serialize $0 \leq -1$ while a mathematical point satisfies $c - 10^{-9} \leq F < c$. Therefore it is numerical closure, not strict infeasibility of every arbitrarily small improver. Its local cutoff discrepancy does not supply a bound on accumulation of all native arithmetic errors; $10^{-9} < 10^{-7}$ alone proves no such global guarantee. This manuscript changes neither guard nor tolerance. No new contradiction with the stated numerical evidence policy was found; claims of error-free rational certification are excluded.

Every numerical original-problem certificate additionally requires a physically verified own U, qualified complete source/return chains and cover, normal lifecycle completion, and the post-seal signed-gap test −1e−7≤U−L≤1e−7. Appendix B.4.1 fully distinguishes that policy, the one-sided native function and the own-zero/source-reduced branches. Tiny signed gaps are reported unchanged. A same-domain counterexample revokes its source and dependent pruning; an independent fleet can remain valid, but floor zero or a good U cannot repair invalid proof dependencies. Source-reduced results are separately labelled and get no dependent nonzero certificate credit.

The single total process deadline guarantees finite execution of this implementation, not certification within every finite window. The finite physical space and ideal complete search do not imply a linear forecast from the remaining gap.

A.8 Writer and variable map

Appendix group	Frozen source authority / mapping
Data, travel, physical objective	Parser.cpp; Evaluator.cpp; Result.cpp physical objective evaluator around line 109
Variables and VTypes	CplexBaseline.cpp 890-1005; $a \leftrightarrow mName$, $Y \leftrightarrow yName$, $\ell \leftrightarrow lName$, $H \leftrightarrow hName$, $Z \leftrightarrow zprodName$
A1-A4	Same writer 1600-1695 and 1807-1824; ConnectivityFlow.cpp downstream-count embedding and Round20Current bounds
A5-A9	Same writer 1826-1932; no Round66 arc-load replacement or Round92 handling activation
A10-A11	Same writer 2241-2274 and 2329-2335
A13-A16	Same writer 2975-3154; stateName and stateGName give s, q; bitName and prodName give plain bits/products
F0 configuration and domains	PaperK1AmSf.cpp 38-114; IntervalRowFactory.cpp 391-807; writer 1697-1750, 2049-2108, 2203-2238, 2311-2327
Factory global/local metadata	IntervalRowFactory.cpp 402-451; verified-incumbent “Global” remains cutoff-scoped
M-B mode isolation	include/Round98StateService.hpp; no projected direction, linked direction, ENS-Q or other opt-in mode
AM, targets, cover and lifecycle	PaperExternalGiniTree.cpp; GurobiBaseline.cpp; ProcessPhaseLedger.cpp; MipStartMapping.cpp

The source is public at the fixed implementation snapshot. The finite production_identity authority records all 216 bindings. Line locations aid review; they do not replace derivations. This appendix supplies constraint groups and validity premises without claiming every engineering identifier as a mathematical contribution.

Appendix B. Executable method specification

This supplement defines the frozen M-B controller and its finite feasible startup H. Algorithm 1 and its two complete subprocedures are in manuscript Section 5; this appendix supplies their data, numerical decisions, decoding and return semantics. Mathematical symbols refer to Appendix A. Historical names are source-map aliases, not additional algorithm variants. The definitions and proofs are retained from R125. Round126 executes only the fixed complete M-B and cold P-GRB entries on the declared new panel; it changes no algorithm.

B.1 Active entry and parameters

The saved R123 M-B command selects `gcap-frontier`, preset `research-round83-vds-equal-net-exchange`, `round98-state-service=m-binary`, `round90-lp-g-split=false`, candidate mode off, witness audit on and H verified-zero stop on. The preset recursively selects joint-seeded interroute decoded descent, VD-P and verified Start; the canonical paper configuration supplies the static row package and first-class interval controller. R122 uses this same M-B method with the panel's declared limits/Seeds. Input, lambda, capacities and physical times remain data, not tuned choices in this round.

Item	Frozen active value / interpretation
Initial geometry	One true-G interval $[0, \min(U, (n-1)/n)]$, n>1; n=1 has G=0
Split	Midpoint; two closed children; depth <8 and width $>10^{-4}+10^{-12}$
Score	Balanced normalized closure, threshold .08; exact formula B.3
Infeasible child policy	Two-child split, then mark proved children empty; contraction_ enabled=false
Native work	Gurobi 13.0.2; one thread; Seed from arm; PresolveAuto; requested MIP gaps 0
Numerical policy	Scheduler epsilon 10^{-7}; FeasibilityTol/OptimalityTol 10^{-6}, IntFeasTol 10^{-5}
Cutoff	Current own U; writer epsilon=0; ObjCon=0
Resource	One process deadline; 30 s shutdown reserve in the saved runs; preparation and H are charged
Products	VD-P; continuous p/d; binary direction retained; A/B on; integer loads retained
Row generation	Static physical plus shared interval factory; no dynamic user/lazy cut separation; native Gurobi cuts remain available
Domain option vs execution	Requested iterative/2; operative one budget intersection plus reachability intersection, recomputed by shared factory; no feedback iteration
H orders	24 shuffled chromosomes from mt19937 seed 20260626; appended constructive order; no genetic generations
H decoder	Full compaction mode 1; cache limit 200000 complete decoded orders; no generation/stagnation quota in this active branch

The "interval-mip-core-no-exhaustive-subset-duration" policy is overridden by the VD-P preset to `round55-vd-p`, still excluding exhaustive subset duration. Compiler identity is the frozen MSYS2 UCRT64 g++ binary SHA256 `8cd0aa578687e4182532222c8734439b3f34ac78b6ce23cba1071c5838025fc1`. Library shuffle and equivalent-key sort order are implementation-version details; the paper does not invent a canonical cross-library tie rule absent in the source.

The unique paper entry is fully resolved in paper_method_config.json; argv_freeze.json contains the exact six native M-B argument vectors and their independent cold controls. PE and DLL SHA256 bind every preset/default value that has no separate CLI option. H seed 20260626 and all native settings are explicit configuration values, not instance-dependent choices. The template below is one token vector (line breaks are presentation only);

INPUT, OUT, physical T, producer B and B_MINUS_6 take exactly their fixed table values. The existing supervisor starts the complete production clock before admission, invokes this vector, and seals normal output; direct native Runtime alone is not T_cert.

```
PE --input INPUT --lambda 0.15 --T T
   --pickup-time 60 --drop-time 60 --time-limit B_MINUS_6
   --process-wall-time-limit B --process-shutdown-margin 30
   --threads 1 --mip-threads 1 --gurobi-seed 0 --gurobi-presolve -1
   --method gcap-frontier
   --algorithm-preset research-round83-vds-equal-net-exchange
   --round90-lp-g-split false --round61-candidate-mode off
   --round65-witness-audit true --round65-hga-zero-stop true
   --out OUT/result.json --log OUT/native.log
   --process-phase-ledger OUT/phases.csv
   --external-gini-artifact-dir OUT/external
   --primal-heuristic-generation-log OUT/hga.csv
   --progress-log OUT/progress.csv --native-evidence-dir OUT/journal
   --round98-state-service m-binary
```

B.2 State, identity and order

A coverage unit has id, interval endpoints, parent id, child index, depth, base/inherited bound, current bound and sources, cutoff, status, and replacement flag. Active statuses are Open (Invalid may remain selectable as an unresolved obligation); Closed, Empty and Fathomed retain their proof role. Replaced parents are represented by their committed children. TerminalReady/Coalesced are dormant sibling-coalescing states, not normal M-B actions. Native status is not interchangeable with coverage status.

Runtime state separately contains canonical artifact path/SHA/row signature, generation and incumbent epoch; LP result (terminal-valid, optimal/infeasible, bound available, raw bound), LP epoch/SHA/endpoints; cached speculative child objects; terminal-started state; native phase count; and the NEXT milestone bit. Own witness state contains complete routes, integer operations, inventory, recomputed U and true G. The model's epigraph G never replaces the latter. Epoch starts at zero and increments on an independently verified incumbent improvement exceeding 10^{-9}. Callback observations and physical witnesses retain separate identities.

Selection uses the **current own U** as cutoff, not an old source cutoff. It considers unreplaced Open/Invalid units with α<U−epsilon and b<U−epsilon. Find their minimum bound and form the epsilon plateau $b_I \leq b_{min} + \epsilon$. Sort the plateau by decreasing width, increasing lower endpoint, increasing upper endpoint, then lexicographic id (the first three comparisons use 10^{-12}). Cycle through this order. Reset the cursor when the plateau membership/order changes or its minimum changes by more than epsilon; after a full cycle increment the tie round. Effort, elapsed time and requested_quantum fields do not affect this `selectNextByBoundOnly` call.

On a new epoch, before reusing a selected artifact verify its old saved bytes, discard its retained backend model, clear artifact/LP state, terminal flags and child cache, and build a new generation at the current cutoff. Qualified old-domain lower bounds remain on their covering units: a smaller cutoff does not invalidate a bound already completed by its old cutoff complement. The NEXT milestone bit is initialized false in each newly created runtime and set true only by a returned NEXT target. **It is not reset in the active epoch invalidation block.** Child-target returns do not set it. No stronger child bound is sent upward to the parent merely because the child has a higher value.

Parent LP reuse requires complete evidence and the same incumbent epoch. A retained artifact must exist and match its SHA. The ordinary cached-child reuse gate requires the parent's ready flag, exactly two cached children and complete child runtime LPs; epoch invalidation clears that flag before reuse. The more elaborate endpoint/path/SHA `validRound90LpGCachedChild` check is conditional on round90 and is off here; its existence is not claimed as an active gate. When reusing children, their stored bounds inherit max(old child bound,current

parent bound) and their cutoff is refreshed. AM still reads the **raw LP bound fields**, not those inherited maxima. Missing/incomplete cache falls back to constructing and solving both children; it is never interpreted as infeasible.

B.3 NEXT, child target and terminal decisions

Call NEXT after the selected parent has completed its LP and merged any qualified bound; re-read the current bound b. The function first rejects nonfinite bounds, and first requeues if any other relevant bound satisfies $b_j + \epsilon < b$, even if the milestone was already reached. Otherwise, if the milestone is false and any $b_j > b + \epsilon$, return NEXT with $t = \min\{b_j : b_j > b + \epsilon\}$. If there is no such bound, or the milestone is true, allow lookahead. The caller supplies only other **Open/Invalid schedulable units** whose current bound is below current U−epsilon (and whose interval remains relevant). Closed, Empty, Fathomed and replaced units are excluded from NEXT competitors even though their floors remain in proof coverage. Thus a lone closed unit at .40 does not supply a NEXT target to an open unit at .20. Source-cutoff metadata governs bound scope, never this current scheduling filter. The function does not use a timed quantum.

If geometry is ineligible, proceed to terminal MIP after NEXT permits lookahead. If eligible, form closed midpoint children in index order 0,1. Each is speculative until the complete replacement transaction. Solve both complete LP relaxations, or use the current admissible cache. A terminal valid child means either proved infeasible, or optimal with an available finite bound; TimeLimit/UNKNOWN is not terminal valid. Deadline exits retain the parent; an unexpected nonterminal/invalid LP is an engineering failure with the parent retained, not permission to prune it. The optional UNKNOWN→core-due research branch is off.

If either qualified child LP is infeasible, AM immediately requests a two-child atomic split. With contraction=false this applies also when both are infeasible; only after installing both does the controller mark the infeasible child or children Empty and discard those backend models. The inactive one-child contraction/both-child parent-close code is not part of M-B.

For two finite optimal LPs with raw bounds $\ell 0$,$\ell 1$, use

$$\Delta = \max(U - b, \max(\epsilon, 10^{-12})), \quad v_j = \min(1, \max(0, (\ell_j - b)/\Delta)), \quad \eta = \min(v_0, v_1), \quad \mu = (v_0 + v_1)/2. \tag{B1}$$

$$\begin{aligned} s &= \max(1, |b|, |\ell_0|, |\ell_1|, |U|), \\ \delta_b &= \max(\epsilon, 32\epsilon_{mach} s), \\ \delta_{score} &= \min(1, \delta_b/\Delta + 32\epsilon_{mach}), \quad \epsilon_{mach} = 2^{-52}. \end{aligned} \tag{B2}$$

Put $b_+ = \min(\ell_0, \ell_1)$. If $b_+ \le b + \epsilon$, return terminal; else if $\eta\mu + \delta_{score} \ge .08$, split; else request a **parent-domain MIP** with bound target b_+. A child target is an intended stopping threshold, not a bound to merge into the parent before native evidence returns. A returned target requeues the parent and re-evaluates the current rule; it does not force a delayed split. This distinguishes NEXT (competitors), child target (lookahead), and terminal (complete parent MIP without a bound-target stop). The recorded adaptive rho is $\min(1, .08/\max(\mu, \max(\epsilon, 10^{-12})))$; it is diagnostic, not another active score rule.

B.4 Native requests, Starts and returned actions

An LP request relaxes the actual original VTypes, keeps the complete row set, and retains the model when permitted. A MIP request restores the exact saved VTypes before optimization. Targets and terminal requests receive the remaining common deadline after preparation, with native TimeLimit set/read back; no new per-leaf budget

is created. Native bound callbacks can request termination when the qualified target is reached. Such a request is not the native return itself.

For enabled verified Start, normalize the current own fleet only among equal-capacity vehicle classes, stable sorting by decreasing number of operations and assigning smaller vehicle ids first. Clear explicit previous Starts even if the replacement is ineligible. Reverify routes, true G and F. Check interval membership with 10^{-8} and non-strict cutoff with $10^{-8}\max(1,|F|,|c|)$. An outside-interval/cutoff fleet is simply ineligible, and the MIP can run without it. Other mapping failures fail the request's engineering gate; they do not establish infeasibility.

For each used route set selected arcs to one, visit z=1, direction to pickup indicator, quantities to its integers, load to each prefix and rank to position. Flow along a q-station route is q,q−1,…,1,0. All absent route columns, including unused vehicles, are zero. Set (Y,r,e,H,G,Z) to physical values, state selector to one at Y, state_g to true G there and zero elsewhere, W_SP=SP, and the unused extrema to actual min/max ratios. Check every actual column bound/type, every actual linear row at 10^{-7}, and actual objective including ObjCon at 10^{-7}; require minimization and no unexpected nonlinear structure. Then submit a complete Start and read it back. Submission, native-log acceptance, integer-vector observation and a later improved incumbent are different observations.

The outcome object is consumed only after the optimize call returns and request finalization detaches callbacks/settles native data. Retaining the same in-memory model can preserve useful solver state, but this paper does not promise that the entire native branch-and-bound tree survives type changes or later calls. Once returned, merge finite qualified bounds only on the actual interval/cutoff; replay a proposed fleet before accepting an improved own U. This order precedes the next controller transaction.

Returned condition after required identity, finalization, zero-gap-setting and consistency gates	Coverage action
LP optimal with finite qualified bound	Merge into that parent, or save on speculative child; child alone changes no active cover
LP infeasible	Parent Empty, or speculative infeasibility retained until atomic split
Target MIP OPTIMAL with bound / INFEASIBLE	Close selected cutoff obligation; retain cutoff-complement floor; discard its model and speculative child models
Target reached flag, termination-request flag, available bound ≥target−epsilon	Merge returned bound; set milestone only for NEXT; requeue; preserve parent and siblings
MIP interrupted/TimeLimit before target	Keep qualified partial bound if engineering gates passed; leave region open and end the common lifecycle
Terminal MIP OPTIMAL / INFEASIBLE	Close its own cutoff obligation; never certify siblings by this status
Terminal MIP interrupted	Leave region open and stop
Unsupported status, missing optimal bound, failed identity, incomplete LP or native error	Stop with explicit failure; preserve unresolved coverage; dependent unqualified evidence is not admitted
No launch because deadline expired	Leave region and previous evidence unchanged; finalize open result

The implementation merges same-unit bounds with max(old bound,new bound), without clipping to U. Empty/Fathomed stores max(previous bound,cutoff); Closed keeps its qualified returned floor. The raw returned L is the scheduler's complete-cover minimum, also **without clipping to U**. Tiny negative signed gaps can therefore occur and remain recorded. On the empty relevant-set path the scheduler helper alone returns 0; the enclosing lifecycle/closure logic determines the outcome.

For the exact mathematical proof, each cutoff-bound source has unconditional regional floor min(bound,source cutoff), and the analytic complement contributes its cutoff floor (A.6). This is proof accounting, not an extra min-assignment performed by the frozen scheduler. Raw numerical evidence and exact proof-accounting expressions must not be interchanged. A.7's numerical closure exception still applies.

B.4.1 Certificate admission: native function and sealed evidence

The native certificate function requires valid complete coverage, all relevant leaves closed, all used bounds certified, monotone bound history, a verified own finite U, completed lifecycle, and source consistency. It then accepts the one-sided test L+max(0,tolerance)≥U. It does **not** enforce a lower bound on the signed gap U−L. In particular, a negative gap of −2e−7 is not rejected by that predicate alone.

The published post-seal reader applies the stronger admission policy. Each native source must have a finished request with matching interval, epoch, model bytes/row signature and parentage; the saved parameters must match the frozen original model (including requested zero relative/absolute MIP gap); every used bound must be finite, available and sourced to the corresponding returned native request, without a counterexample or consistency failure. It reconstructs the represented children and their full interval union, preserves unresolved obligations, requires every relevant unit closed for certification, replays the own physical fleet and objective, and requires normal process completion (no killed, timed-out producer, missing seal, or failed engineering gate). These gates are conjunctions, not substitutes for one another. A clean nonzero certificate additionally requires −1e−7≤U−L≤1e−7; gaps are never clipped. Thus the above −2e−7 case is rejected after sealing.

The clean independent own-zero branch requires a physically replayed own U≤1e−12, nonnegative lambda and weights, and **no started native request**; it returns L=0 and a synthetic complete zero cover. The startup early-stop test is 0≤verified F≤1e−12; the neutral physical loop itself stops only on exact F=0. If native sources are damaged, the source-reduced fallback is stricter: it needs every Y_i=D_i and recomputed F=G=P=0 exactly. Its independent L=0 does not restore any invalid native source or dependent nonzero certificate. A retained positive feasible fleet alone supplies U, never certificate credit for damaged proof dependencies.

These are distinct functions in the frozen evidence chain: ExternalGiniTree.cpp:460-489 for native admission; read_round122/123 → read_round118.item/finalize → read_round116.reconstruct_arm → round108_reader.endpoint (328-391), with CLOSURE_TOL=1e−7 and ZERO_TOL=1e−12; round114_claims/round114_audit supplies damaged-source fallback. The rules are stated here so those links are audit references, not missing algorithm definitions.

B.5 Teaching states and a separately sourced event

These first numbers are **synthetic teaching states**, not measurements. Take U=1, epsilon=10^{-7}. With selected b=.20 and competitor bounds (.20,.35,.60), the plateau contains the tied .20 units and NEXT=.35. After a qualified NEXT return b=.35 the milestone is true; if all other relevant bounds are at least .35, lookahead is permitted. (If the tied .20 unit still exists, the requeue check wins first.)

For a fresh controlling parent b=.20 with no strict competitor, children (.21,.30) give Δ=.8, v=(.0125,.125), score=.000859375, b+=.21: run parent MIP to .21. Children (.20,.90) instead give b+=b: terminal, regardless of the strong sibling. Children (.50,.60) give v=(.375,.5), score=.1640625: split, install both children, with union floor .50. If the other covering unit has floor .25, global L remains .25. A target TimeLimit at .205 keeps the parent open at its qualified .205 bound; a missing Start changes none of these implications.

The following is an **actual R123 saved event** from M-B's `adaptive_mass_decision_ledger.csv` and `paper_tree_events.csv`, selected as a finite trace excerpt. On L0.0.0=[0,.09308544387476185], depth 2, b=.33371261877766062, raw child LPs (.34112380854511071,.35882784229299913), U=.3723417754990474 give eta=.19185481632186216, mu=.42100858578645295, score=.08077252489598688 and score tolerance=.000002588718179786521 6. The decision is split; the event at telemetry 127.2749697 records atomic replacement and global floor .34112380854511071. Earlier recorded splits at 38.3552841 and 72.829994 used the infeasible-child two-child branch. These times are native-run telemetry, not newly measured clocks or a reconstructed unknown checkpoint. Compact source excerpts and hashes accompany the finite checks.

B.6 H encoding and top-level procedure

Physical routes are labelled lists $0, i_1, \ldots, i_q, 0$ with one signed integer operation $a_i = p_i - d_i \neq 0$ per station; missing vehicle means empty route. H's **order encoding** is different: M ordered lists partition all n stations, including a tail that the decoder may not execute. A chromosome is a permutation of station ids 1..n and M−1 identical separator tokens; splitting at separators preserves empty vehicle lists. Reordering an encoded tail does not itself describe a feasible service plan. Decoding returns signed operations; materialization deletes zero-operation stations, closes used routes at depot and verifies the entire fleet.

```
H(instance, common deadline):
    C := empty physical fleet, verify Y=b
    repeat choose and admit best joint insertion (B.7) until none/deadline
    keep verified C as an own candidate
    generate all 24 shuffled order chromosomes with mt19937(20260626)
    append C's order after deterministic tail completion (below)
    initialize a shared complete-decode cache, empty
    decode all 25 orders in order, publishing verified improvements
    for each seed in order:
        repeat generate guided candidates (B.10), order by proxy fitness
        fully decode candidates in that order
        accept first fitness improvement >1e-12; restart this seed's pass
        if every candidate checked without improvement: finish this seed
        honor verified-zero/deadline between calls and candidates
    take the best independently verified candidate, including C
    run strict insertion/quantity closure followed by best neutral block move;
        repeat these two phases (B.11) until zero/no neutral/deadline/failure
    return best admitted complete fleet and stop reason
```

Constructive tail completion starts with C's visited order and each vehicle's complete physical duration as a proxy. For each unvisited station i in increasing id, append it to the vehicle minimizing proxy[k]+tau(last,i)+tau(i,0)−tau(last,0), choosing the first vehicle on an exact tie; update that proxy. The remaining operations are decided by the same full decoder. Twenty-four shuffles are generated before appending this extra seed, preserving random draws. `constructive_mode=0` means there are no additional biased population members. The snapshot's C++ shuffle is authoritative for exact seed orders; the algorithm specifies the generator/seed/order and permits cross-library sequence differences. HGA crossover, mutation, education generations, 2000-generation stagnation and obsolete "10 iterations" fields do not run in this decoded-descent branch.

The outer physical candidate store keeps a candidate only after full verification and an improvement greater than 10^{-10}, unless no incumbent exists. Verified-zero can terminate startup early. This publication threshold differs from inner 10^{-12} move thresholds; an internally accepted tiny improvement need not replace the outer witness.

B.7 Joint insertion

At a verified fleet let $L_k(a)$ be load on leg a, numbering the original legs from 0 through the number of visited stations. For each ordered (pickup station i or 0,drop station j or 0), excluding (0,0), i=j, and already visited nonzero stations, enumerate vehicles and positions. Zero denotes an absent side, not depot stock. Both sides use the same integer quantity q. For a pair choose original legs a≤b, inserting i then j; on one leg the sequence is predecessor,i,j,successor. Travel increment is the sum of replaced arc differences (one combined expression when a=b).

The load upper amount is $Q_k - \max_{v=a..b} L_k(v)$ for a pair; $Q_k - \max_{v=a..end} L_k(v)$ for pickup only; $\min_{v=a..end} L_k(v)$ for drop only. Intersect with available inventory at i, room at j and max vehicle capacity. Complete duration increment is travel increment plus hq when a pickup is inserted, and just travel increment

for drop only (that drop replaces return unloading). The time test uses T+10^{-7}. For each amount, retain the cheapest feasible travel placement, ties (vehicle,a,b), by a suffix scan of placement amount limits. Evaluate all amounts 1..limit using F on the changed one/two inventories; accept only gain>10^{-12}.

Choose globally: nonpositive added duration before positive; then greatest gain per added duration (compare cross products, or gain for both nonpositive), greater gain, smaller added duration, and lexicographically (i,j,q,k,a,b). Apply and fully replay; require objective agreement within 10^{-10} and verified gain>10^{-12}. Every insertion adds at least one previously unvisited station, so construction has at most n accepted insertions. This finite placement/amount scan is a heuristic neighborhood, not an optimization backend.

B.8 Single and paired quantity modification

Only currently served stations participate. Enumerate a increasing, then b=0 (single) and each served b>a (pair). For every nonzero integer t set $a'_a = a_a + t$, and $a'_b = a_b - t$ if b exists. Station bounds give

$$b_a - C_a - a_a \le t \le b_a - a_a; \qquad a_b - b_b \le t \le a_b - b_b + C_b \quad (b \neq 0). \tag{B3}$$

Intersect these with every affected prefix load $0 \le L + \chi t \le Q_k$, where chi is +1 after a, −1 after b, zero before/beyond the cancelling pair. This includes cross-vehicle modifications; each route has its own capacity. Enumerate all integers in the resulting interval increasing, skip zero. Convert signed operations to pickup=max(a',0), drop=max(−a',0), remove zero stops and shortcut travel. Sign reversal is allowed. Check complete duration, including depot unloading, at 10^{-7}; evaluate F; require gain>10^{-12}. Choose lowest proposed objective, tie (a,b,t). Full replay and objective agreement within 10^{-10} precede acceptance. Missing stations cannot be added by this operator; insertion handles them.

Strict closure computes both the best insertion and best quantity move on the same current fleet, chooses smaller proposed F, insertion on an exact tie, admits the verified strict improvement, and repeats. If neither exists, that union neighborhood is exhausted. Verification failure retains the previous fleet and records failure. This is exactly the Round76 subprocedure called by active exchange descent; the standalone Round75/76 preset entry branches are not separately run by main.

B.9 Complete greedy decoder and its compensating moves

For each encoded route take its maximal prefix whose travel with direct return stays within T+10^{-12}, stopping at the first exceeding station. Fix this travel during that decoder pass. Set the cumulative pickup budget $B_k = \max(0, \lfloor (T - travel + 10^{-12})/h \rfloor)$ (an integer sentinel max-int/4 if $h \le 10^{-12}$); this is a handling budget, not Q. Initialize all operations zero and Y=b. The single-route proxy version instead keeps other-route operations fixed. The active mode is 1: visit vehicles and stations forward, greedily match each current surplus above clipped target to later deficits, as much as remaining supply, demand, station bounds, all intervening load bounds and pickup budget allow; leftover surplus may be collected for depot return. This constructive initialization does not require each step to decrease F.

Define an interval move by vehicle k, positions a<b, sign $\sigma \in \{+1, -1\}$, amount $q \ge 1$. Increase a's signed operation by sigma q and decrease b's by sigma q. The sink may instead be the depot after the last position, in which case only a changes. Loads on [a,b) change by sigma q. Each final inventory must stay in [0,C], loads in [0,Q], and total positive operations in [0,B_k]. Thus an earlier delivery can be reduced/reversed and a pickup compensated elsewhere; there is still only one net one-way operation per physical station. Feasible amount bounds come from station room and the minimum/maximum load on the changed interval; the positive-part pickup change is piecewise linear with breakpoints at operation sign changes. The implementation finds the

largest feasible q by these segments, then evaluates every positive integer up to it. The feasible amount sublevel contains q=0 and is an interval, so no integer hole is skipped. Objective arithmetic updates the changed ratios, pairwise absolute sum and penalties; it is not an LP/DP solve.

The active decoder loop is:

```
repeat:
    repeatedly apply the best strict atomic interval move, until none
    if a strict same-route composite exists: apply best; restart
    if a strict two-route composite exists: apply best; restart
    if a strict unit interval polish exists: apply best; restart
    if a strict flow-pair redistribution exists: apply best; restart
    if no atomic move occurred in this pass: return current operations
```

Atomic enumeration is vehicle, source position, later sink positions then depot, sign + then −, amount increasing. Keep the best move and the eight best atoms per route. To update the best, initialize value F−EPS and replace only when candidate+EPS<best, EPS=10^{-12}; this gives an effective initial gain test >2EPS. For each bounded list, append candidates until size K ($K = 8$ atoms or $K = 6$ local candidates), maintaining a max heap with comparator a.objective<b.objective. Once full, replace its front only if candidate.objective+EPS<front.objective, then restore the heap. Final sorting compares only ascending objective, with no secondary key. Equivalent-key heap positions consequently depend on the frozen standard library. Local candidates comprise each retained atom and each pair of distinct atoms on one route, adding their operation deltas simultaneously. Recheck all combined loads, positive-operation budget and station bounds, then retain at most six best candidates per route using the same bounded-heap objective logic. Candidates can individually worsen F: their combination may compensate that worsening. Same-route best selection uses F−EPS and candidate+EPS<best. Cross-route candidates combine one retained local candidate from each pair of vehicles, recompute the full common Gini/penalty objective for their jointly changed (up to eight) stations, and use the same strict test. Distinct routes have disjoint stations, so combined physical feasibility follows from their checked local feasibility; joint objective evaluation remains necessary.

Unit polish checks amount 1 for every legal atomic interval, with initial best F−10^{-14} and candidate+10^{-14}<best. Flow-pair polish ignores zero operations, partitions the route into consecutive same-sign runs, and pairs each pickup run with its immediately following drop run. For each ordered pair of distinct such run pairs, choose one pickup and drop position in each: subtract one unit from the first pickup/drop flow and add one to the second. Check both interval load changes together, station bounds and original signs; pickup total is unchanged. Its candidate helper still requires gain>10^{-12}, followed by the 10^{-14} best-comparison guard. Residual pair-unit, route-exact and pair-exact neighborhoods return false in this active implementation; they are not hidden optimization oracles.

After a first full decode, if its executed prefix contains zero-operation stations, delete them, keep only its nonzero active order, and run this same full decoder from zero operations **once more** on those compacted routes. Return whichever has smaller F, first decode on exact tie. This does not repeat compaction to a fixed point and does not restore the removed tail. The cached object contains complete signed operations and fitness=−F. Materializing that object removes remaining zero operations and independently checks the original physical route duration.

Decoder cache key is compaction mode plus the ordered lists with separators; it includes tail positions, even if they decode identically. On a hit return the entire stored decode. On a miss at size≥200000 clear the whole map before inserting; this is neither LRU nor a permanent visited-state ban. Internal atomic/local caches are tied to an incremented operation-state epoch and cleared after accepted updates. The static ratio/penalty preprocessing cache keys capacities, targets and weights (including source slice offsets); it affects arithmetic reuse, not candidate admission. No general accepted-state de-duplication is claimed.

B.10 Guided order descent and ties

For each route compute the travel-only executable prefix as above. Within it record first negative-operation position and last positive-operation position. A station's role is pickup if its decoded operation>0, drop if <0; if zero, use surplus/shortage of decoded final inventory relative to target, otherwise neutral. Supply anchor is first drop, or position 0; demand anchor is immediately after last pickup, otherwise prefix end.

Generate, in vehicle order: move the last zero-operation prefix station to that route's end; collect every nonneutral prefix position plus the first supply and first demand in its tail. Move a supply lying after first drop, or in tail, to the supply anchor; move a demand before last pickup, or in tail with a last pickup present, to the demand anchor. Duplicate generated candidates are not globally removed. Same-route erase/insert decrements destination position if it originally lay after the erased position; skip unchanged order. For interroute proposals take only the first supply/demand in each source tail, each different destination in order: insert supply at destination's supply anchor if that route contains any demand-role station; insert demand at its demand anchor if it has an executed pickup.

The proxy fully runs the single-route incremental decoder with other-route operations fixed. For a two-route change try both route decoding orders and take the larger fitness (smaller F). Proxies only order proposals: sort decreasing proxy fitness, then fully decode in that order and accept the first fitness increase>10^{-12}. An unsuccessful pass evaluates every generated proposal, including a poor proxy. There is no top-k rejection here.

The guided comparator has no secondary key: equal proxy values are equivalent to C++ `std::sort`, which is not stable. Likewise internal bounded-heap/objective-only sorts have implementation-dependent equivalent-key order. For the frozen compiler/library they supply the recorded run's repeatable behavior; a cross-language implementation may choose a different equivalent-key order and consequently a different heuristic path. Specifying lexicographic ties here would change the frozen algorithm. By contrast B.7/B.8 and the physical block operators do have explicit deterministic ties. This limitation affects path-level reproduction, not feasibility, exact model scope or native lower-bound validity.

B.11 Neutral block moves and exit argument

Strict closure is followed by one best neutral move from the union below. The potential is the **complete** vector of all vehicle durations, sorted decreasingly, including zeros for unused vehicles; compare it with exact C++ lexicographic <, never by an aggregate sum or a tolerance that ignores an earlier increase.

Balanced relocation enumerates source vehicle, all nonempty contiguous half-open position blocks [a,b), other target vehicle and every insertion leg. Require net signed operation zero. If relative prefix load extrema inside the block are lo,hi, require target load+lo≥0 and target load+hi≤Q_target. Removing a net-zero block leaves source suffix loads unchanged; inserting it leaves target suffix loads unchanged. Keep operation quantities and order inside the block; recompute both full durations and accept only if each≤T+10^{-7} and the sorted vector strictly improves. Choose smallest potential, tie (source,a,b,target,leg).

Equal-net exchange enumerates every contiguous nonempty block per route, bucketed by integer net operation in increasing net order; pair vehicles s<t and blocks from equal-net buckets. At each insertion prefix require its old load plus the incoming block's relative extrema in that vehicle's [0,Q]. Equal net preserves all suffix loads, even when the common net is nonzero. Replace the two blocks in their original internal order, preserve all station operations, recompute complete durations and compare the same potential. Choose smallest potential, tie $kind, source, a, b, target, target_a, target_b$, with kind 0 relocation before kind 1 exchange. `bestRound83Neutral` evaluates the whole relocation neighborhood first and then exchanges.

Before accepting a neutral candidate, independent replay requires identical final-inventory vector, **exactly equal recomputed objective**, and exact equality between recomputed duration potential and the proposed potential,

which must strictly improve the prior one. Failed admission retains the prior fleet. After an accepted neutral move, run strict closure again. Stop on exact verified objective zero, common deadline, failed verification, or closure plus no improving neutral move. A selected no-move event is an exhaustion claim only for these enumerated neighborhoods.

In ideal arithmetic, after the finite construction the accepted physical phases strictly decrease (F,sorted durations) lexicographically, and the finite space of labelled routes/integer quantities excludes cycles. The order-descent phase strictly increases its deterministic decoded fitness on a finite set of orders. The implementation uses incremental floating arithmetic and tolerances in the decoder; these comparisons are not a proof of exact rational monotonicity or an accumulated error bound. The physical block phases have exact inventory/objective equality for neutral steps, unlike a merely tolerance-neutral rule. There is no fabricated de-duplication argument. The common deadline and external lifecycle cap are the operational stopping boundary; expensive decoder calls are finite routines and do not have a separate native optimization timeout. H neighborhood closure never certifies the original optimum.

Complete historical-result display supplement

This display is generated one way from the six unchanged R125 table authorities. Numeric display uses twelve significant figures; the adjacent original CSV/Markdown files retain their exact published bytes and full precision. N is a nonzero numerical certificate; O is a qualified open endpoint. Historical WIN labels retain their old endpoint rule and are not new certification wins.

Table S1a. All 32 R122 physical/bound endpoints

Role	Seed	Method	U	L	Signed gap	Tier
PB-C20-B	0	P-GRB	0.224680311823	0.176578375157	0.0481019366668	O
PB-C20-B	0	ENS-C	0.224680311823	0.224680311823	-1.11022302463e-16	N
PB-C20-B	0	M-B	0.224680311823	0.224680311823	1.05471187339e-15	N
PB-C20-B	0	MB-D1	0.224680311823	0.224680311823	2.49800180541e-16	N
PB-B20-S	0	ENS-C	0.863152320565	0.863152320565	-2.22044604925e-16	N
PB-B20-S	0	M-B	0.863152320565	0.863152320565	-2.22044604925e-16	N
PB-B20-S	0	MB-D1	0.863152320565	0.863152320565	-2.22044604925e-16	N
PB-B20-S	0	P-GRB	0.863152320565	0.863152320565	1.11022302463e-16	N
PB-C35-U	0	M-B	0.169636402195	0.0839518542929	0.0856845479019	O
PB-C35-U	0	MB-D1	0.179300436585	0.08353399353	0.0957664430545	O
PB-C35-U	0	P-GRB	0.174771887556	0.0597029031719	0.115068984384	O
PB-C35-U	0	ENS-C	0.164237487795	0.0815753937702	0.0826620940247	O
PB-B35-B	0	MB-D1	0.234581409857	0.0920079836183	0.142573426238	O
PB-B35-B	0	P-GRB	0.305555778217	0	0.305555778217	O
PB-B35-B	0	ENS-C	0.246504470413	0.13228796803	0.114216502383	O
PB-B35-B	0	M-B	0.246504470413	0.127634301846	0.118870168567	O
PB-C50-S	0	MB-D1	0.372341775499	0.356874739941	0.0154670355577	O
PB-C50-S	0	M-B	0.372341775499	0.356874739941	0.0154670355577	O
PB-C50-S	0	ENS-C	0.372341775499	0.357980799883	0.014360975616	O
PB-C50-S	0	P-GRB	0.425950075294	0.349220695801	0.0767293794932	O
PB-B50-U	0	P-GRB	1.17695575854	0.0683793824136	1.10857637613	O
PB-B50-U	0	MB-D1	0.677298419001	0.471234869552	0.206063549448	O
PB-B50-U	0	M-B	0.697473509598	0.45534539996	0.242128109638	O
PB-B50-U	0	ENS-C	0.832548015217	0.452360367383	0.380187647834	O
PB-B35-B	1	ENS-C	0.234581409857	0.114641011444	0.119940398412	O
PB-B35-B	1	P-GRB	0.279798434442	2.02718966417e-18	0.279798434442	O
PB-B35-B	1	MB-D1	0.234581409857	0.138322042028	0.0962593678281	O
PB-B35-B	1	M-B	0.246504470413	0.0982245517985	0.148279918614	O
PB-C50-S	1	M-B	0.372341775499	0.357008333973	0.0153334415264	O
PB-C50-S	1	ENS-C	0.372341775499	0.358161762014	0.0141800134847	O
PB-C50-S	1	P-GRB	0.391780558579	0.349450859646	0.0423296989334	O
PB-C50-S	1	MB-D1	0.372341775499	0.357008333973	0.0153334415264	O

Table S1b. All 32 R122 complete clocks

Role	Seed	Method	Cap	T_produce	T_verify	T_total
PB-C20-B	0	P-GRB	900	872.589019559	1.23015520003	873.819174759
PB-C20-B	0	ENS-C	900	430.5575901	3.21318039999	433.7707705
PB-C20-B	0	M-B	900	395.8595413	3.24970759999	399.1092489
PB-C20-B	0	MB-D1	900	665.0687819	4.85072250001	669.9195044
PB-B20-S	0	ENS-C	900	4.62243311576	1.44725659999	6.06968971575
PB-B20-S	0	M-B	900	2.6319677	1.57636520002	4.20833290002
PB-B20-S	0	MB-D1	900	2.6983887	2.1314255	4.8298142
PB-B20-S	0	P-GRB	900	116.0327756	1.02186559999	117.0546412
PB-C35-U	0	M-B	2400	2372.52643333	8.951676	2381.47810933
PB-C35-U	0	MB-D1	2400	2370.6203998	11.0386199	2381.6590197
PB-C35-U	0	P-GRB	2400	2370.6161682	1.09987460001	2371.7160428
PB-C35-U	0	ENS-C	2400	2370.5989279	8.8968437	2379.4957716
PB-B35-B	0	MB-D1	2400	2372.44035804	10.1910831	2382.63144114
PB-B35-B	0	P-GRB	2400	2370.5561489	0.518086399999	2371.0742353
PB-B35-B	0	ENS-C	2400	2370.552666	9.1507964	2379.7034624
PB-B35-B	0	M-B	2400	2371.7861588	8.1583865	2379.9445453
PB-C50-S	0	MB-D1	3600	3572.88123229	40.0557249	3612.93695719
PB-C50-S	0	M-B	3600	3570.5953431	40.3517955	3610.9471386
PB-C50-S	0	ENS-C	3600	3571.8953901	40.0828359	3611.978226
PB-C50-S	0	P-GRB	3600	3583.0201592	1.57247429999	3584.5926335
PB-B50-U	0	P-GRB	3600	3572.99104926	1.30260140001	3574.29365066
PB-B50-U	0	MB-D1	3600	3570.5630142	25.6511639	3596.2141781
PB-B50-U	0	M-B	3600	3571.8623909	19.9168553	3591.7792462
PB-B50-U	0	ENS-C	3600	3570.5730882	19.7308828	3590.303971
PB-B35-B	1	ENS-C	2400	2372.97340551	18.6435541	2391.61695961
PB-B35-B	1	P-GRB	2400	2370.5116857	1.97982149999	2372.4915072
PB-B35-B	1	MB-D1	2400	2370.6168373	22.4221823	2393.0390196
PB-B35-B	1	M-B	2400	2370.4895614	18.640514	2389.1300754
PB-C50-S	1	M-B	3600	3572.60278401	85.3058211	3657.90860511
PB-C50-S	1	ENS-C	3600	3570.6597372	84.4460346	3655.1057718
PB-C50-S	1	P-GRB	3600	3570.672291	930.1450991	4500.8173901
PB-C50-S	1	MB-D1	3600	3572.56418137	85.1722841	3657.73646547

Table S2. All 48 R122 directed comparisons

Role	Seed	Candidate	Control	Class	Severe
PB-C20-B	0	MB-D1	P-GRB	WIN	False
PB-C20-B	0	M-B	P-GRB	WIN	False
PB-C20-B	0	ENS-C	P-GRB	WIN	False
PB-C20-B	0	MB-D1	M-B	LOSS	False
PB-C20-B	0	MB-D1	ENS-C	LOSS	False
PB-C20-B	0	M-B	ENS-C	TIE	False
PB-B20-S	0	MB-D1	P-GRB	WIN	False
PB-B20-S	0	M-B	P-GRB	WIN	False
PB-B20-S	0	ENS-C	P-GRB	WIN	False
PB-B20-S	0	MB-D1	M-B	TIE	False
PB-B20-S	0	MB-D1	ENS-C	TIE	False
PB-B20-S	0	M-B	ENS-C	TIE	False
PB-C35-U	0	MB-D1	P-GRB	MIXED	False
PB-C35-U	0	M-B	P-GRB	WIN	False
PB-C35-U	0	ENS-C	P-GRB	WIN	False
PB-C35-U	0	MB-D1	M-B	LOSS	False
PB-C35-U	0	MB-D1	ENS-C	LOSS	False
PB-C35-U	0	M-B	ENS-C	LOSS	False
PB-B35-B	0	MB-D1	P-GRB	WIN	False
PB-B35-B	0	M-B	P-GRB	WIN	False
PB-B35-B	0	ENS-C	P-GRB	WIN	False
PB-B35-B	0	MB-D1	M-B	MIXED	False
PB-B35-B	0	MB-D1	ENS-C	MIXED	False
PB-B35-B	0	M-B	ENS-C	TIE	False
PB-C50-S	0	MB-D1	P-GRB	WIN	False
PB-C50-S	0	M-B	P-GRB	WIN	False
PB-C50-S	0	ENS-C	P-GRB	WIN	False
PB-C50-S	0	MB-D1	M-B	TIE	False
PB-C50-S	0	MB-D1	ENS-C	TIE	False
PB-C50-S	0	M-B	ENS-C	TIE	False
PB-B50-U	0	MB-D1	P-GRB	WIN	False
PB-B50-U	0	M-B	P-GRB	WIN	False
PB-B50-U	0	ENS-C	P-GRB	WIN	False
PB-B50-U	0	MB-D1	M-B	WIN	False
PB-B50-U	0	MB-D1	ENS-C	WIN	False
PB-B50-U	0	M-B	ENS-C	WIN	False
PB-B35-B	1	MB-D1	P-GRB	WIN	False
PB-B35-B	1	M-B	P-GRB	WIN	False
PB-B35-B	1	ENS-C	P-GRB	WIN	False
PB-B35-B	1	MB-D1	M-B	WIN	False

Role	Seed	Candidate	Control	Class	Severe
PB-B35-B	1	MB-D1	ENS-C	WIN	False
PB-B35-B	1	M-B	ENS-C	LOSS	False
PB-C50-S	1	MB-D1	P-GRB	WIN	False
PB-C50-S	1	M-B	P-GRB	WIN	False
PB-C50-S	1	ENS-C	P-GRB	WIN	False
PB-C50-S	1	MB-D1	M-B	TIE	False
PB-C50-S	1	MB-D1	ENS-C	TIE	False
PB-C50-S	1	M-B	ENS-C	TIE	False

Table S3a. All three known R123 endpoints

Method	U	L	Signed gap	Tier
P-GRB	0.37764170855	0.350377933041	0.0272637755094	O
M-B	0.372341775499	0.358011485552	0.0143302899472	O
ENS-C	0.372341775499	0.358943666168	0.0133981093313	O

Table S3b. All three R123 complete clocks

Method	Cap	T_produce	T_verify	T_total
P-GRB	18000	17971.7112989	1.96611939999	17973.6774183
M-B	18000	17970.5725766	40.964555	18011.5371316
ENS-C	18000	17970.8751446	58.2420274	18029.117172

Table S4. All three R123 directed comparisons

Candidate	Control	Class	Severe
M-B	P-GRB	WIN	False
ENS-C	P-GRB	WIN	False
M-B	ENS-C	TIE	False

Table S5: existing checkpoints

Method	Time (s)	U/last witness	L	Source
P-GRB	3600	0.425950075294	0.349220695801	last_prior_existing_committed_records
P-GRB	7200	0.42083291094	0.349949886971	last_prior_existing_committed_records
P-GRB	10800	0.389701007874	0.350050746873	last_prior_existing_committed_records
P-GRB	14400	0.378043868561	0.350180030064	last_prior_existing_committed_records
P-GRB	end 17971.711298929702	0.37764170855	0.350377933041	actual_sealed_endpoint_before_checkpoint
M-B	3600	0.372341775499	0.356874739941	last_prior_existing_committed_records
M-B	7200	0.372341775499	0.357080008664	last_prior_existing_committed_records
M-B	10800	0.372341775499	0.357681585834	last_prior_existing_committed_records
M-B	14400	0.372341775499	0.357815221734	last_prior_existing_committed_records
M-B	end 17970.572576599981	0.372341775499	0.358011485552	actual_sealed_endpoint_before_checkpoint
ENS-C	3600	unknown	unknown	last_prior_existing_committed_records
ENS-C	7200	unknown	unknown	last_prior_existing_committed_records

Method	Time (s)	U/last witness	L	Source
ENS-C	10800	unknown	unknown	last_prior_existing_committed_records
ENS-C	14400	unknown	unknown	last_prior_existing_committed_records
ENS-C	end 17970.875144599995	0.372341775499	0.358943666168	actual_sealed_endpoint_before_checkpoint